UNDERGRADUATE RESEARCH OPPORTUNITY PROGRAMME (UROP) PROJECT REPORT

From Social Signals to Market Reactions: Novel Information versus Social Amplification in Polymarket

Leakage-Safe Forecasting of Prediction-Market Repricing: News Signals and a Replicated Quote-Change Forecaster

By

Zhu Rong

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Advisor: Lu Yao

Deliverables:

Report: 1 Volume
Program: EventX source repository (Python), frozen release manifests and evaluation scripts
Data: hash-locked evaluation outputs and receipts (data/ directory)

Abstract

EventX asks whether information available at a decision time predicts short-horizon repricing in prediction markets, under strict point-in-time and evaluate-once rules. In a frozen study of 4,651 Kalshi and Polymarket markets, adding counts of conservatively available news articles to a market-history model failed all five promotion gates. A prospective test of richer news features, run after a disclosed revision of its data audit, also did not improve on market history. We then defined a narrower target: the probability that a contract's quoted midpoint 30 minutes later differs from its current value. A frozen gradient-boosted classifier on 18 point-in-time quote features reduced log loss against a logistic baseline by 9.2–16.8% on three unopened blocks, two of them with never-used contracts, with contract-clustered intervals above zero. Post hoc, each contract's own online outcome rate came close to the booster on new contracts. A frozen logistic combination of the booster, that rate and the contract's outcome-history length then beat the rate alone in log loss by 13.3% on 51 further new contracts (29 June–12 July) and by 10.5% on 78 more (13–21 July), so pooled quote information added value beyond each contract's own history. The July test also confirmed two prespecified segments: the booster's gain over the rate was concentrated after recent quote moves and at wide spreads (Holm-adjusted $p = 0.0004$ for both). A stack that also used these two signals then beat the earlier stack on 52 more contracts over 22 July–8 August, a reused holdout period (log loss 0.3006 against 0.3083; interval [0.002, 0.014] nats), 14.0% below the rate. Six recent ICML, NeurIPS and ICLR learners, run with default settings, did not beat these models. In development, decayed counts of recent quote changes lowered held-out log loss by a further 4.0%, a candidate for a new test. A three-day live test issued 68.4% of scheduled forecasts on time, with log loss 16.7% below the logistic baseline on known outcomes (descriptive). In these data, short-horizon quote activity was predictable from market state, while news features did not help.

Subject Descriptors:

G.3 Probability and Statistics
I.2.6 Learning
H.2.8 Database Applications (Data mining)
J.4 Social and Behavioral Sciences (Economics)

Keywords:

Prediction markets, probabilistic forecasting, market microstructure, gradient boosting, leakage-safe evaluation

Implementation Software and Hardware:

Python 3.12, scikit-learn, PyTorch, PyArrow, NumPy, httpx; Polymarket CLOB WebSocket and Gamma API; pmxt hourly order-book archive; Lumid FinData; Apple macOS laptop (17 GB RAM)

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1 Introduction

1.1 Background and motivation

Prediction markets let traders buy contracts that pay one dollar if an event happens, so a contract's price can be read as a crowd probability of that event (Wolfers and Zitzewitz, 2004; Arrow et al, 2008). Platforms such as Polymarket and Kalshi now list thousands of contracts on elections, sports, macroeconomic releases, technology and geopolitics. Prices move when new information arrives, but they also move for mechanical reasons: liquidity providers re-quote, the order book thins out overnight, and contracts become quiet as their resolution approaches.

The original goal of EventX was a benchmark for whether outside information (news articles, and later posts by influential social-media accounts) carries **incremental, point-in-time** predictive information about these price moves, beyond what market history already reveals. Results in this area are easy to overstate. A single look-ahead leak, a reused test period or a weak baseline can produce an apparent gain without any error showing up. The project therefore treats evaluation discipline as a first-class research output. The intended protocol is that every test set is frozen before it is opened, every model is frozen before it is scored, and each held-out period is evaluated exactly once; the exceptions that occurred are disclosed in Appendix B.

1.2 Research questions

1. **RQ1 (news increment).** Does news that was demonstrably available at decision time improve a forecast of the next 30-minute price jump beyond a market-history baseline?
2. **RQ2 (forecastable target).** Is there a short-horizon market quantity that can be forecast reliably better than strong baselines, and does that advantage survive new time periods and never-seen contracts?
3. **RQ3 (live operation).** Can such a forecaster run prospectively on live data, under the same frozen rules, with every decision auditable after the fact?

1.3 Summary of contributions

- A frozen, preregistered historical test of news value over 4,651 markets, with a negative answer to RQ1. Under the frozen design, news counts made forecasts slightly worse and failed every promotion gate. A post hoc market-clustered bootstrap places the

upper bound of the effect below every positive effect size registered before labels were seen (Chapter 4).

- A leakage-safe, evaluate-once framework: protocols and cohorts frozen before data are read, consumption receipts, hash-locked releases, preserved failures, and a checker that ties each registered headline number to its artifact (Chapters 3 and 5, Appendix C).
- A quote-change forecaster whose advantage over logistic, rate-based and earlier boosted baselines replicates on two unopened June weeks, including 35 contracts never used before. Post hoc, a contract's own online outcome rate scores close to it on the new contracts; quote features appear most useful when little contract-specific outcome history is available, and their value after history accumulates remains uncertain on unseen contracts. In a descriptive breakdown its log loss is lower in every market category (Chapters 5 and 6). A third unopened block, 21 further new contracts on 15–28 June, met the same frozen rule (Section 6.5).
- A confirmatory test of the question those post hoc results raised. A logistic stack of the booster, the contract's online rate and its outcome-history length, frozen on opened blocks, beat the rate alone by 13.3% in log loss on 51 new contracts over 29 June–12 July, with a contract × date interval of [0.034, 0.071] nats (Section 6.6). It passed again on 78 further new contracts over 13–21 July, where two prespecified segment hypotheses were also confirmed (Section 6.8).
- A segmented account of where the forecasting gain lies. A post hoc analysis over five blocks found the booster's advantage over a contract's own rate concentrated just after quote moves and at wide spreads, with no category or time-of-day pattern; a frozen test on 78 new contracts confirmed both effects (Holm-adjusted $p = 0.0004$). In those active states, 21% of rows, the stack's log loss was 15.7% below the contract's rate. A stack that uses the two segment signals, developed on opened blocks and tested once on 52 further new contracts, improved on the earlier stack (Sections 6.7–6.9).
- A comparison with recent state-of-the-art baselines from ICML, NeurIPS and ICLR 2024–2025 (TabM, RealMLP, TabICL, Moirai-MoE, Sundial, Time-MoE), run with default settings on matched inputs; none beat the project's boosters or stacks, and a development study identifies recent change-history features as the next candidate (Sections 6.10 and 6.11).

- A prospective runtime that validates its own clock against network time servers, writes hash-chained journals, and refuses to evaluate more than once. A three-day live test on 30 newly selected contracts issued 68.4% of its scheduled forecasts on time; on known outcomes its log loss was 16.7% below the logistic baseline, a descriptive result (Chapter 7).
- A full record of failed attempts and protocol deviations, kept rather than hidden (Appendix B).

1.4 Organisation of the report

Chapter 2 reviews related work. Chapter 3 describes the data and the leakage safeguards shared by all studies. Chapter 4 reports the news-increment study (Track A). Chapters 5 and 6 present the quote-change forecaster and its historical evaluations (Track B), and Chapter 7 the prospective live test. Chapter 8 discusses validity, and Chapter 9 concludes with limitations and recommendations. Appendices give feature definitions, the deviation log and reproduction instructions.

2 Background and Related Work

2.1 Prediction markets as forecasts

Prediction-market prices aggregate dispersed beliefs and often forecast at least as well as polls or experts (Wolfers and Zitzewitz, 2004; Arrow et al, 2008). Calibration is good on average but imperfect, especially for long-shot outcomes and far-off events (Page and Clemen, 2013). This project does not try to beat the market's forecast of the final outcome. It forecasts the behaviour of the market's *own price* over the next half hour.

2.2 News, attention and asset prices

Media content has measurable, short-lived associations with stock returns and trading volume (Tetlock, 2007), so news is a natural candidate signal for prediction markets as well. The difficulty is timing. A news record is only useful if it was available to a trader *before* the price moved, so timestamps must reflect availability (publication, ingestion and feed-delivery times) rather than event time. EventX therefore uses the most conservative available timestamp, and treats missing feed intervals as unknown rather than as zero news.

Prediction-market and betting prices are known to respond to news quickly. Snowberg, Wolfers and Zitzewitz (2007) use intraday moves in election-night prediction-market prices to measure how expected election outcomes moved financial markets. Croxson and Reade (2014) find that soccer betting prices incorporate goals scored just before half-time quickly and fully. Diercks, Katz and Wright (2026) study Kalshi macroeconomic markets as a high-frequency benchmark of expectations and examine how they respond to macroeconomic and financial news. These studies concern prices reacting to identified events. Track A asks a different, harder question: whether routinely available news counts, measured before the decision time, improve a forecast beyond what market history already contains.

2.3 Market microstructure

Short-horizon price behaviour is driven largely by microstructure: the bid–ask spread, quote updates by liquidity providers, and the arrival of orders (O'Hara, 1995; Hasbrouck, 2007). The midpoint of the best bid and ask is the standard high-frequency price proxy. Whether a midpoint changes over a short window depends heavily on how active the book has recently been. Activity is persistent, which makes a quote-change target forecastable in principle. Engle and Russell (1998) model the clustering of durations between transactions. A duration model predicts whether at least one update arrives within a window, which is related to but not the

same as the target here: a midpoint can move and return, so an update does not guarantee that the endpoint differs.

Machine-learning work on limit order books mostly forecasts the *direction* of the next mid-price move in liquid equity markets. Ntakaris et al (2018) provide a benchmark dataset with a day-based anchored cross-validation protocol; Zhang, Zohren and Roberts (2019) apply convolutional and recurrent networks to order-book data; and Sirignano and Cont (2019) report that a model pooled across stocks predicts the direction of price moves out of sample, including for stocks outside its training sample. EventX differs in target, market and protocol. It forecasts whether a prediction-market midpoint differs after a fixed 30-minute horizon, a coarser activity target, in thin markets whose contracts expire; and it tests a model frozen before the data were retrieved, once, on contracts never used before. Its closest analogue to the cross-stock result of Sirignano and Cont is the 8–14 June test on new contracts, so transfer to unseen instruments is not itself new. Kolm, Turiel and Westray (2023) find, across 115 stocks and several horizons, that the choice of input representation (order-flow imbalance rather than raw book levels) matters more than network depth. This suggests that input representation and horizon, rather than model size, are the natural next variables for EventX.

2.4 Forecast evaluation and leakage

Probabilistic forecasts are evaluated with strictly proper scoring rules, which reward honest probabilities: here, log loss and the Brier score (Brier, 1950; Gneiting and Raftery, 2007). Boosted trees can produce systematically distorted probability estimates (Niculescu-Mizil and Caruana, 2005), so calibration is checked separately. Leakage, meaning information from the future or from the test set entering model building, is a common cause of false results in data mining (Kaufman et al, 2012), and a survey of machine-learning-based science found it in published work across 17 fields (Kapoor and Narayanan, 2023). In financial time series it also arises from overlapping label windows, which requires purged, embargoed cross-validation (López de Prado, 2018). Observations within one market are dependent, so uncertainty is estimated by resampling whole blocks or whole contracts rather than individual rows (Künsch, 1989; Cameron, Gelbach and Miller, 2008).

2.5 Learning method

The forecasters use histogram-based gradient boosting (Friedman, 2001; Ke et al, 2017), as implemented in scikit-learn (Pedregosa et al, 2011). EventX makes no claim of algorithmic novelty. Its contribution is the target definition, the point-in-time feature construction, and the

evaluation protocol around a standard learner. Tree ensembles remain highly competitive on medium-sized tabular data when the comparison accounts for tuning (Grinsztajn, Oyallon and Varoquaux, 2022), so boosting is a reasonable default. The same benchmark shows why a fair comparison needs matched inputs and tuning budgets, which the primary comparison here does not provide (Section 5.4).

2.6 Recent state-of-the-art learners used as baselines

Older sequence networks (recurrent and convolutional models such as DeepLOB) and plain multilayer perceptrons are no longer state-of-the-art reference points, so the baselines in Section 6.10 come from ICML, NeurIPS and ICLR papers of the last two years. Recency alone does not make a comparison fair or a method suitable for this task, so each baseline is run with its published default configuration on matched inputs, and the departures from matched tuning are stated with the results. For tabular prediction, TabM (Gorishniy, Kotelnikov and Babenko, 2025) makes a multilayer perceptron behave like an efficient ensemble by sharing most weights across members; RealMLP (Holzmüller, Grinsztajn and Steinwart, 2024) is a multilayer perceptron with tuned defaults that matches boosted trees on medium-sized benchmarks; and TabICL (Qu et al, 2025) is a tabular foundation model that predicts by in-context learning from the training rows, with no gradient training on the target task. The TabArena living benchmark (Erickson et al, 2025) finds that boosted trees remain strong on practical tabular data, that tuned and ensembled deep models have caught up, and that foundation models excel on small training sets, which is the regime of row_hgb7 (9,212 training rows). For a contract's own outcome history, the natural modern reference is a pretrained time-series foundation model; Moirai-MoE (Liu et al, 2025) adds sparse mixture-of-experts layers to the Moirai forecaster and is used zero-shot. Sundial (Liu et al, 2025b) and Time-MoE (Shi et al, 2025) are equally recent; their released code runs as remote code from the model hub, so it was downloaded, read and hash-locked before use (Appendix B). All three foundation models forecast real-valued series, so applying them to a binary outcome series is an adaptation (Section 6.10), and a poor score evaluates that adaptation rather than the models in general. Work on large language models that forecast the final outcomes of prediction-market questions (Halawi et al, 2024; Karger et al, 2025) addresses a different target, the event outcome rather than the next half hour of the quote, and so does not provide a direct baseline.

3 Data and Leakage Safeguards

3.1 Data sources

- **Lumid FinData** supplies historical executed trades for Kalshi and Polymarket, and a news feed with publication and ingestion timestamps. All ingestion runs as resumable, checkpointed batch jobs under the provider's rate limit, and analysis uses frozen, content-hashed extracts rather than live API calls.
- **The pmxt Polymarket archive** publishes one Parquet file of order-book events per hour (book snapshots, price changes, trades), each with venue and receive timestamps. Files are sorted by market identifier, so the project can download only the row groups that contain the selected contracts.
- **Polymarket's Gamma metadata API and CLOB WebSocket** supply contract metadata (question, outcomes, token identifiers, start and end dates), and live book updates for the prospective test (Polymarket, 2026).

3.2 Safeguards common to all studies

The following rules govern the studies in this report and are enforced in code rather than by convention. The quote freshness and validity rule applies to Track B only.

- **Point in time.** Every feature for a decision at time t uses only records received at or before t. In Track B, a forecast is issued only when the latest quote is fresh (at most 120 seconds old on both the venue and the receive clocks), valid (bid below ask, spread at most 0.05) and unambiguous.
- **Frozen, hash-locked releases.** Protocols, cohorts, model files and evaluation code are frozen as SHA-256 manifests *before* the data they will be judged on is retrieved. A changed file means a new version, never an edit in place.
- **Evaluate once.** Each test window is consumed by a single evaluation run, which writes a consumption receipt first and refuses to run again. Results are reported whether favourable or not.
- **Protected identifiers and windows.** 1,231 market identifiers and three time windows used by earlier confirmatory studies are filtered out at decode time, so no Track B experiment can read them.

- **Outcome-blind cohort selection.** New contract cohorts are chosen from identity, event-type and clock columns plus metadata only, before any price or quote column of the test period is read.
- **Missing outcomes stay missing.** If the 30-minute endpoint quote is unavailable, the forecast is scored as unknown. Deterministic worst-case bounds over all issued forecasts are reported alongside the scored results.

3.3 Study overview

The project ran a sequence of versioned studies. Table 1 lists each one with its data, model, comparator and status, so that the version names used in later chapters can be looked up in one place. Training periods are those of the frozen model being scored; evaluation periods are the data that model had not been fitted on.

Table 1. Overview of the studies reported here. Known / missing counts are matched forecasts with an observed endpoint and issued forecasts without one.

Study	Training data	Evaluation data	Model vs comparator	Contracts; known / missing	Pass rule	Status
V3.1 development (Track A)	Purged expanding folds	17 May–22 Jul 2026, out-of-fold	B1 (news counts) vs B0 (market history)	2,290 of 4,651 markets contribute; 1,460,454 rows	All five gates (Section 4.1)	Complete: 0 of 5 gates; B1 not promoted
V3.1 holdout	17 May–22 Jul	22 Jul–8 Aug	B0 only (absolute benchmark)	300 markets; 211,718 rows	None	Consumed once; deviation disclosed
V4 confirmatory	17 May–22 Jul (discovery)	6 Sep–30 Oct, collected prospectively	B3 (richer news) vs B0	470 contracts	LL gain ≥ 0.001 nats; bootstrap lower bound > 0; Brier and venue guards	Fail (after disclosed audit revision)
May 13–19 replication (Track B)	7–9 May, 50 development contracts	13–19 May	row_hgb7 vs activity logistic	30; 27,924 / 4,553	None (fixed-model replication)	Complete

Study	Training data	Evaluation data	Model vs comparator	Contracts; known / missing	Pass rule	Status
June 1–7	7–9 May	1–7 June, unopened	row_hgb7 vs activity logistic	18 (development); 16,453 / 2,851	Lower LL than activity logistic and global rate; Brier no higher	Complete: passed
June 8–14	7–9 May	8–14 June, unopened, new contracts	row_hgb7 vs activity logistic	35; 35,218 / 4,438	As June 1–7, plus contract-clustered lower bound > 0	Complete: passed; DNS refetch deviation
June 8–14, V03 secondary	7–9 May (14 features)	8–14 June	V03 hgb7 vs its full logistic	35; 35,749 / 4,268	Registered secondary analysis	Complete
June 15–28	7–9 May	15–28 June, unopened, new contracts	row_hgb7 vs activity logistic; V03 secondary	21 selected	As June 8–14, for both models	Pass (both models)
History stack (new cycle)	Four opened blocks, 13 May–28 June	29 June–12 July, unopened, new contracts	stack_lin vs online contract rate	53 selected, 51 contribute; 65,458 / 8,739	Lower LL and Brier no higher than the rate; contract × date lower bound > 0	Pass
Segment confirmation (new cycle)	Five opened blocks, 13 May–12 July	13–21 July, unopened, new contracts	row_hgb7 gain over online rate by segment (H1, H2); stack vs rate (H3)	80 selected, 78 contribute; 159,283 / 4,535	One-sided contract-clustered p, Holm over two; H3 as history stack	Pass (all three)
Segment-aware stack (new cycle)	Six opened blocks, 13 May–21 July	22 July–8 August, new contracts (disclosed deviation: V3.1 holdout calendar period)	seg_hgb vs stack_lin	54 selected, 52 contribute; 134,341 / 6,632	Lower LL and Brier no higher than stack_lin; contract × date lower bound > 0	Pass

Study	Training data	Evaluation data	Model vs comparator	Contracts; known / missing	Pass rule	Status
Recent baselines (post hoc)	As each reference model	Seven opened blocks, 13 May–8 August	TabM, RealMLP, TabICL, CatBoost vs row_hgb7; Moirai-MoE, Sundial, Time-MoE vs online rate; end-to-end networks vs seg_hgb	285 contracts; 506,723 rows	None (descriptive, scored once)	Complete
Development study v8	Leave-one-block-out over seven opened blocks	Held-out block of each fold	Change-history and online-offset boosters vs seg_hgb design	285 contracts; 506,723 rows	None (development)	Complete; e2e_hx the next test candidate
Live V02b	7–9 May (V03)	27 Oct 06:35–29 Oct 23:25 UTC, prospective	V03 hgb7 vs full logistic and global rate	30; 22,650 scheduled decisions	None; reporting floor of 100 known outcomes and 20 of each class	Floor met; descriptive

3.4 Scores, uncertainty and missing outcomes

All forecasts are probabilities p of a binary outcome y. They are scored with two strictly proper scoring rules, for which lower is better:

- **log loss**, the mean of $-[y \ln p + (1 - y) \ln(1 - p)]$ over forecasts, in nats;
- **Brier score**, the mean of $(p - y)^2$.

Average precision (AP) and the area under the ROC curve (AUC) are reported as ranking summaries. A **gain** is always the comparator's score minus the candidate's score, so a positive gain favours the candidate; for AP, which is higher-is-better, the change is candidate minus comparator. A **relative reduction** is the log-loss gain divided by the comparator's log loss: a 16.8% reduction means the booster's log loss is 16.8% below the comparator's. Comparisons are **matched**: every model is scored on exactly the same forecasts.

Uncertainty. Forecasts from the same contract are dependent, so intervals come from a **cluster bootstrap**. Whole contracts (or, in Track A, whole markets) are resampled with replacement, the row-weighted mean gain is recomputed for each resample, and the 2.5th and 97.5th percentiles of 5,000 resamples (2,000 in Section 4.3) form the 95% interval. A date-clustered version resamples whole days instead. The one-sided 95% upper bound used in Section 4.3 is the 95th percentile.

Missing outcomes. A forecast is issued from the quote at time t, but its outcome is known only if a valid endpoint quote exists at $t + 30$ minutes. Forecasts without one are scored as unknown, never imputed. The **missing-outcome bounds** are deterministic, not confidence intervals: each unknown outcome is set to whichever value is worst (lower bound) or best (upper bound) for the candidate relative to the comparator, and the mean gain is recomputed over all issued forecasts. A lower bound above zero means that no assignment of the unknown outcomes could reverse the sign of the gain.

4 Track A: Incremental Value of News

4.1 Design of the historical primary study (V3.1)

The cohort contains 4,651 markets (3,832 Kalshi and 819 Polymarket) with a unique canonical YES outcome, at least 100 positive-notional executions and at least ten active days. Development covers 17 May to 22 July 2026, and a separate holdout covers 22 July to 8 August 2026. At each five-minute decision time the price is the latest executed YES trade. The binary target asks whether the absolute log-odds change over the next 30 minutes exceeds four times the trailing one-minute volatility.

B0 is a 12-feature market-history logistic ensemble: price log-odds, 5/30/120-minute momentum, 30/240-minute realised volatility, 60/240-minute trade counts and notional, time since last trade, and venue. **B1** adds rolling counts of conservatively available, lexically associated news articles over 60, 360 and 1,440 minutes. Both are trained on identical rows across five purged, expanding-time folds. B1 would be promoted only if all five gates held: positive aggregate average-precision (AP) and Brier changes, positive changes in at least four folds, a positive lower bound on a 5,000-replicate paired block-bootstrap Brier interval, and no venue-level deterioration.

4.2 Results

Table 2. V3.1 development comparison of news-augmented (B1) and market-only (B0) models on 1,460,454 matched out-of-fold rows.

Quantity	Value	Interpretation
AP change (B1 − B0)	−0.001916	news worse
Brier improvement, Brier(B0) − Brier(B1)	−0.000284	news worse
Paired block-bootstrap Brier interval	[−0.000341, −0.000230]	excludes zero, against news
Folds improving (AP / Brier)	0 of 5	gate failed
Promotion gates passed	0 of 5	B0 selected

All five gates failed, so, as preregistered, only B0 was evaluated on the holdout: AP 0.171628 and Brier 0.107107 on 211,718 rows from 300 markets. That number is an absolute market-only benchmark, not an estimate of news value. One protocol deviation affects it: the holdout builder printed the positive-label count before the consumption receipt was written. Weights and formulas were already frozen, and the deviation is recorded (Appendix B).

4.3 How large a news effect is ruled out

A failed gate shows only that no gain was found. To say how large a gain the data still allow, the 2,290 markets that contribute out-of-fold rows (of the 4,651 in the cohort) were resampled as clusters, 2,000 times, over the frozen five-fold out-of-fold predictions. This is a post hoc, descriptive analysis of already-opened development data: the consumed holdout is not read, nothing is refit, and the point estimates reproduce the stored ones exactly.

Table 3. Market-clustered bootstrap of the news increment on 1,460,454 development out-of-fold rows from 2,290 markets. Improvements are B0 minus B1 for Brier and log loss; the AP change is B1 minus B0. Positive values would favour news.

Quantity	Point	95% interval	One-sided 95% upper bound
Brier improvement	−0.000284	[−0.000395, −0.000179]	−0.000196
Average-precision change	−0.00192	[−0.00333, −0.00052]	−0.00075
Log-loss improvement (nats)	−0.00098	[−0.00137, −0.00059]	−0.00066

The one-sided 95% upper bound of the Brier improvement is below zero, conditional on this cohort and resampling design. Markets that share events or dates may be dependent, so the interval may be somewhat too narrow. For scale, B0's Brier skill over climatology is 0.00210 (base rate 14.1%), and the upper bound corresponds to about 9% of that skill; this is a scale comparison, not a confidence interval for the ratio. Before any label was read, a frozen design-sensitivity simulation estimated how often a true effect would pass all five gates: 7% for a "very small" effect (Brier +0.0001, AP +0.0005), 33% for "small", 86% for "moderate" (Brier +0.0005, AP +0.0025) and 99.8% for "clear". Every registered scenario, down to "very small", lies above the observed upper bounds (Figure 1). The null is therefore informative about the effect sizes the study was designed to detect, although it does not show that news has no value in general. The bound applies to lexically associated article counts over 60–1,440 minutes. Richer representations are the subject of V4.

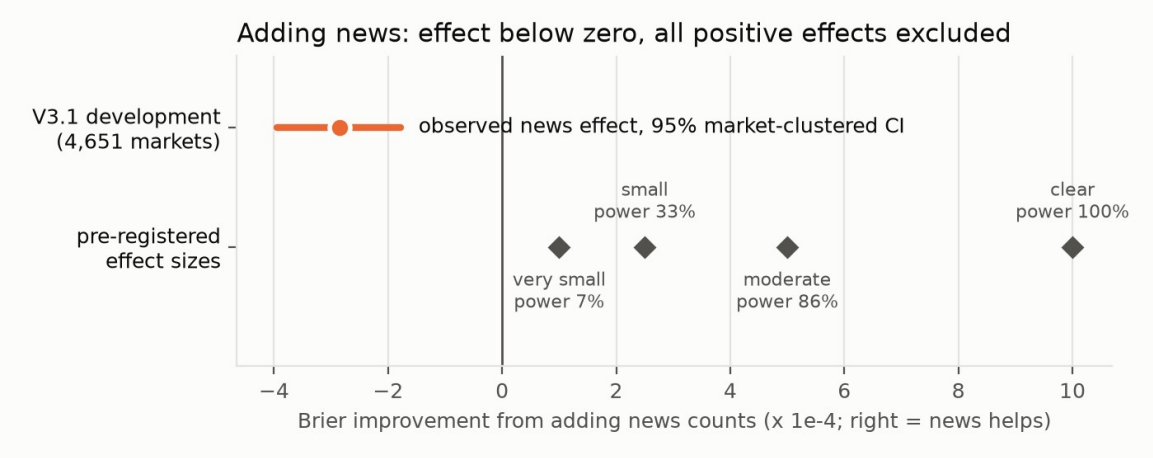


Figure 1. Observed news effect on Brier score with its 95% market-clustered interval, against the effect sizes registered before labels were read and their simulated probabilities of passing all five gates.

4.4 Richer news features (V4) and diagnostics (V5, V6)

V4 freezes a nested ladder of richer news features: news volume (B1); relevance and seven-day novelty (B2); and publisher diversity with cross-publisher headline diffusion (B3). It tests them prospectively on 470 Polymarket contracts from 6 September to 30 October 2026. Success requires at least 0.001 nats of log-loss improvement for B3 over B0, with a positive bootstrap lower bound and Brier and venue guards. The sealed collection passed a label-blind health check on 27 October: 58 of 58 news days and 51 of 51 trade days had receipts, nothing was pending, and all 15 frozen hashes matched. At close-out on 30 October the frozen label-blind audit, the first step of the chain, stopped on the first news partition. It required every article's upstream ingestion-run status to be `ok`, but only 74,119 of the 186,753 sealed article rows (40%) have that status, and every day has others (partial 90,630, completed 16,220, failed 5,521, running 263). The frozen models were trained on discovery news with the same mix of statuses, and no V4 feature code filters on status, so the rule could never have passed and was inconsistent with the frozen pipeline. Before the holdout was consumed, and before any V4 outcome was read, a versioned audit (v2) was written and hash-locked. It accepts any recorded status and keeps every other check. This is a disclosed deviation from the frozen implementation (Appendix B). The v2 audit passed, and the frozen chain ran once on 31 October, scoring 47,279 rows from 470 contracts in the primary population (Table 4).

Richer news did not help. B3's log loss was 0.0008 nats worse than B0's, with a 95% event-clustered bootstrap interval of [−0.0052, 0.0012] over 245 event clusters. Only the Brier guard was met, so the positive claim is not supported. A caveat limits what this says about news: every model in the ladder transferred poorly to this period, with AUC about 0.40 (below chance) and

a mean forecast of 0.17 against an observed rate of 0.085. The test therefore shows that richer news did not rescue a market-history model that failed to transfer, rather than giving a clean estimate of the value of news. The simpler news-volume and relevance models (B1 and B2) improved log loss by 0.004 and 0.003 nats, but these are secondary contrasts, not the confirmatory one.

Table 4. V4 confirmatory result, primary population (price age at most 30 minutes). Improvements are B0 minus B3, so positive favours B3. 47,279 rows from 470 contracts; evaluated once on 31 October after the disclosed audit revision.

Quantity	Value	Frozen condition	Met?
B3 − B0 log-loss improvement (nats per row)	−0.00078	at least 0.001	No
95% bootstrap interval of the improvement	[−0.0052, 0.0012]	lower bound above zero	No
B3 − B0 Brier improvement	−0.00014	at least −0.0005	Yes
Venue-level estimates (Polymarket)	−0.00078	non-negative where at least 30 clusters	No
Positive claim supported	No	all four conditions	—

Exploratory work on already-opened data (V5) traced the apparent benefit of a residual-news model to controls. A context-only gate improved more than the full news correction did, and a nonlinear market-only model beat the full news model by 0.0085 nats. The log-loss improvement from adding the news mechanism itself was −0.000871 nats, with interval [−0.001457, −0.000208]; that is, it made log loss slightly worse. A contract-aware news-linking effort (V6) failed its strict link audit, and certified predictive support was 0 of 8,844 contract-days. Taken together, the answer to RQ1 so far is **no**: under strict availability rules, simple news representations did not add value beyond market history.

5 Track B: Quote-Change Forecasting Method

5.1 Why the target changed

Before settling on the current target, the project tried to forecast the *exact* midpoint 30 minutes ahead, both historically and in a live audited run on 25 October. Because most midpoints do not change over 30 minutes, the persistence forecast ("no change") is very hard to beat on mean absolute error. In the live run a ridge model was 30.9% worse than persistence on 257 observed outcomes. This does not show that point forecasting is inherently unpromising, only that these point models did not beat persistence. The project therefore switched to a binary target, **whether** the midpoint 30 minutes later will differ from the current one, which is forecast as a probability and scored with proper scoring rules.

5.2 Target and eligibility

At each five-minute grid time t, the target is 1 if the recorded YES midpoint at $t + 30$ minutes differs from the midpoint at t by more than 10^{-12}, and 0 otherwise. The tolerance removes floating-point noise only; it is not an economic threshold. A forecast is issued only when the current quote passes the freshness and validity rules of Section 3.2. A target is assigned only when the endpoint quote passes the same rules. The target measures quote activity. It carries no information about direction, size, final resolution or profit.

Why forecast quote activity. The target is a measure of near-term market activity, not of direction. It is studied here for three reasons, which are motivations rather than demonstrated benefits. First, it resolves every 30 minutes, so a frozen model can be tested on tens of thousands of outcomes within days; this makes it a practical testbed for leakage-safe, evaluate-once protocols. Second, a calibrated activity forecast could plausibly help schedule monitoring, quote refreshes or data collection, and it provides a strong market-only baseline against which news or social signals can be judged. Third, much short-horizon price movement is mechanical, and a study of external information has to control for that predictable component first.

What the target misses. A one-tick move and a large repricing both count as 1, and a move that reverses before the endpoint counts as 0. The target says nothing about the direction of the next move, the resolution of the underlying event, or profit after spreads and fees. Directional forecasting would need a different target and different baselines, and any economic value would need evidence this project does not provide.

5.3 Features

The primary model (**row_hgb7**) uses 18 features computed from quotes received by decision time (Appendix A):

- the midpoint and its log-odds, and the spread;
- the source and receive ages of the quote;
- the hour of day, encoded as sine and cosine;
- the number and fraction of valid quote states in the last 30 minutes;
- signed and absolute midpoint changes over 5, 15 and 30 minutes, each with an explicit missing flag.

The live-runtime model (**V03 hgb7**) uses a 14-feature subset that drops the age and activity-count features. Those four depend on archive receive clocks that a live system cannot reproduce exactly. No news, order-depth or contract-identity features are used, so the models are contract-agnostic.

5.4 Models and baselines

The forecaster is a histogram gradient-boosting classifier (100 iterations, learning rate 0.05, at most 7 leaves, minimum leaf size 100, L2 penalty 10, no early stopping and no class reweighting). Every comparison is against fixed baselines. The rate and logistic baselines were fitted on the same training rows as row_hgb7; the legacy booster was loaded unchanged from an earlier frozen bundle:

- a smoothed global event rate;
- a per-contract rate shrunk toward the global rate with 20 pseudo-observations;
- an **activity logistic regression** on nine features (midpoint, spread, log valid-update count, and absolute 5-, 15- and 30-minute changes with their missing flags), the primary comparator;
- the earlier frozen booster (**legacy HGB**), loaded from its own earlier bundle rather than refit;
- for the V03 live model only, its own full-feature logistic regression on the V03 features.

row_hgb7 was selected and refit on 7–9 May 2026 data from 50 development contracts before any later block was retrieved. Nothing has been refit, recalibrated or re-thresholded since.

The primary comparison is between two complete forecasting systems: row_hgb7 uses 18 inputs and activity logistic regression uses nine of them. It therefore shows that the booster forecasts better than the specified reference, but not why. The gain could come from nonlinearity, from the nine extra inputs, or from both. The V03 comparison against a logistic regression on the same 14 inputs is better controlled, but it is a separate model with its own result and does not repair the primary comparison retroactively.

5.5 Evaluation protocol

Each evaluation block follows the same sequence. First the protocol is frozen, including the primary comparison and the pass rule. Then the cohort is selected outcome-blind, the hourly archive files are captured (with one permitted retry for transport failures), and finally the block is evaluated exactly once. The primary metric is matched, row-weighted log loss of row_hgb7 against activity logistic regression. Each block's pass rule was frozen with its protocol, and the two June rules differ:

- **1–7 June:** on the development cohort, the booster's log loss is below both activity logistic regression and the global rate, and its Brier score is no higher than logistic regression's. A contract-clustered bootstrap interval was added before evaluation as an analysis-only secondary result, not a pass condition;
- **8–14 June:** the same conditions, and in addition the 95% contract-clustered bootstrap interval for the log-loss gain over activity logistic regression must have a lower bound above zero (5,000 resamples of whole contracts).

In both blocks, date-clustered intervals, per-date and per-contract win counts, a nonoverlapping 30-minute subset, a strict continuous-path subset, and worst-case missing-outcome bounds are secondary results. A log-loss or Brier **gain** is always the baseline's score minus the booster's, so positive values favour the booster.

6 Track B: Historical Results

6.1 Development and early replications

An earlier frozen booster cut log loss by 18.1% against activity logistic regression on a new time block, 28–30 April (3,729 rows, 11 markets). The broader-trained row_hgb7 reduced log loss by 22.7% on 10–12 May for the development cohort and by 9.6% for ten reserved contracts. On the seven-day replication of 13–19 May, row_hgb7 achieved 0.3583 against logistic regression's 0.4524 (20.8%). For six training-excluded contracts the gain was only 3.5%, and their worst-case missing-outcome bounds crossed zero. That gap motivated the dedicated test on new contracts in Section 6.3.

6.2 Unopened week, same cohort: 1–7 June

No earlier study had read any June quote data. All 168 hourly files were captured, with 0 protected rows read. Only 20 of the 60 fixed contracts were still quoting, because the rest had expired or gone inactive. These were kept unchanged rather than replaced.

Table 5. Matched results, 1–7 June, development cohort (18 contributing contracts, 16,453 known outcomes, 2,068 positives). Lower log loss and Brier are better.

Model	Log loss	Brier	AUC	Booster log-loss reduction
row_hgb7	**0.2931**	**0.0848**	0.835	—
Activity logistic	0.3523	0.1003	—	16.8%
Legacy HGB	0.3285	—	—	10.8%
Global rate	0.3854	—	0.500	24.0%

All registered pass conditions held. In the secondary, analysis-only bootstrap, the 95% contract-clustered interval for the gain over activity logistic regression is [0.039, 0.080] nats (18 contracts) and the date-clustered interval is [0.047, 0.073]. The robustness results below use all 20 contributing contracts (18,677 known outcomes; log loss 0.2948 against 0.3527): the booster won on 7 of 7 dates and 17 of 20 contracts, and the nonoverlapping subset (0.2922 against 0.3471) and the strict subset (0.1554 against 0.2270) agree. For the development cohort, the worst-case missing-outcome bounds over 19,304 issued forecasts are [−0.001, +0.100] nats: the lower end is slightly negative under the most adverse assignment of the 2,851 unknown endpoints.

6.3 Never-used contracts: 8–14 June

To test transfer to new identities, 40 contracts were selected by metadata only. Their identifiers lie in a hexadecimal range disjoint from every earlier cohort, they are binary Yes/No, at most one is taken per event and none shares an event with earlier cohorts, and they were open across the whole week. They span sports, politics, finance, technology and geopolitics. A local DNS outage failed 74 hourly downloads on both permitted attempts. Before any evaluation, those hours were refetched outcome-blind into a separate directory, and 165 of 168 hours were then available. This exceeds the single-retry rule, so it is recorded as a protocol deviation. The original source is preserved unevaluated.

Table 6. Matched results, 8–14 June, never-used contracts (35 contributing contracts, 35,218 known outcomes, 8,830 positives).

Model	Log loss	Brier	AUC	Booster log-loss reduction
row_hgb7	**0.4208**	**0.1335**	0.827	—
Activity logistic	0.4943	0.1584	0.756	14.9%
Legacy HGB	0.4873	0.1547	0.785	13.6%
Global / contract rate	0.5843	0.1945	0.500	28.0%
V03 hgb7 (live model)	0.4270	0.1358	0.825	19.7% vs its full logistic

All registered pass conditions held, including the positive bootstrap lower bound. The contract-clustered interval for the gain over logistic regression is [0.033, 0.119] nats from 35 clusters. The booster won on 7 of 7 dates and 27 of 35 contracts, and its equal-contract-weighted log loss is 0.4222 against 0.5134. The worst-case missing-outcome bounds over all 39,656 issued forecasts are [+0.009, +0.114] nats, positive even if every unknown endpoint had gone against the booster. A secondary analysis of the V03 live model, registered before evaluation, gave a 19.7% reduction with interval [0.066, 0.147] and wins on 31 of 35 contracts. That cleared V03 for the live test in Chapter 7.

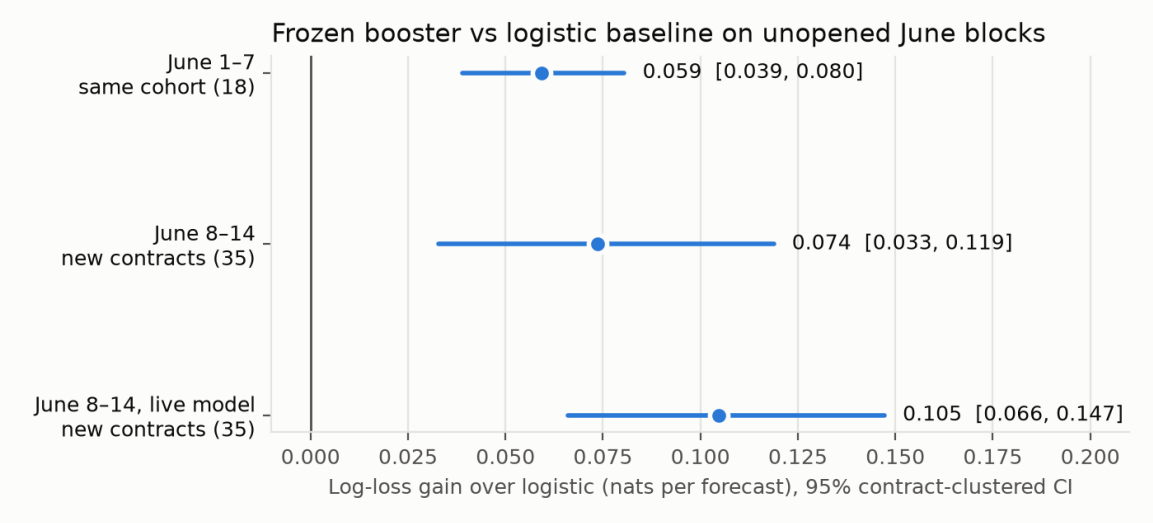


Figure 2. Log-loss gain of the frozen booster over logistic regression on the two unopened June blocks, with 95% contract-clustered bootstrap intervals. Contributing contracts are given in parentheses.

6.4 Calibration, drivers and category breakdown

Figure 3 shows that on 8–14 June both models are close to the diagonal below a forecast probability of 0.2, where most forecasts lie. Both under-predict somewhat in the upper deciles, the booster slightly less at the top. Because the evaluation must not change the frozen model, no recalibration was applied.

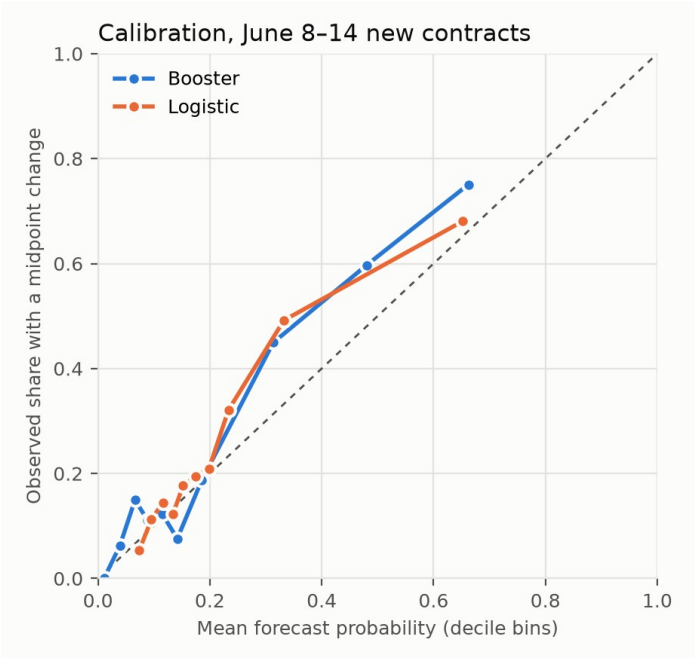


Figure 3. Reliability diagram for the booster and logistic regression on 8–14 June (decile bins of forecast probability).

On 1–7 June (Figure 4) both models over-forecast, which is consistent with a lower change rate in June than in the May training data. The expected calibration error is 0.046 for the booster and

0.052 for logistic regression. This differs from 8–14 June, where both models under-predicted in the upper deciles. The booster's forecasts are also sharper: its decile forecasts span 0.02 to 0.59, against 0.08 to 0.44 for logistic regression. These descriptions do not decompose the score gain into calibration and resolution components. Recalibration could improve both models, but it would need a new frozen version and a new unopened block, since these blocks are consumed.

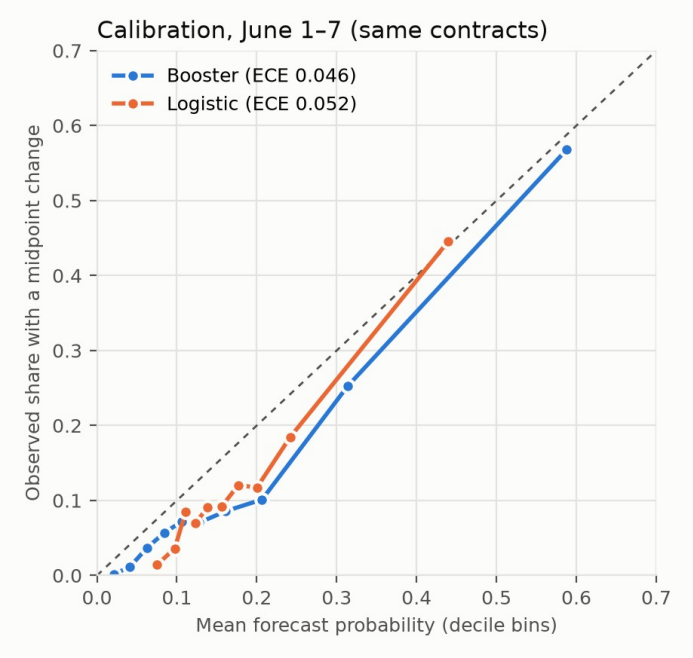


Figure 4. Reliability diagram on 1–7 June, development cohort (decile bins), with equal-frequency expected calibration error.

What the booster relies on. To describe which inputs the fitted booster uses, the frozen booster was re-applied to the consumed 1–7 June grid ledger (reproducing the stored forecasts exactly). Each feature group was then permuted jointly across rows, ten times, and the resulting increase in log loss was recorded (Figure 5). Recent midpoint moves dominate: permuting the 30-, 5- and 15-minute groups raises log loss by 0.021, 0.019 and 0.018 nats. Price level (0.015), update activity (0.008) and spread (0.006) follow. Quote freshness and time of day contribute almost nothing (at most 0.0004). The low weight of the age features is consistent with the V03 live model dropping them. It does not reduce the need for an accurate clock, which still governs input selection, eligibility and emission deadlines. Jointly permuting the nine inputs that only the booster sees raises its log loss by 0.013 nats, 21% of its 0.059-nat advantage over logistic regression, and the perturbed booster still beats logistic regression. This is a sensitivity analysis of a fitted model, not a decomposition of the gap into extra inputs and nonlinearity: permutation also breaks deterministic links between features, such as midpoint and its log-odds, or signed

and absolute changes. Isolating the role of nonlinearity would need a separately designed comparison with matched features. Permutation importance is descriptive and correlated features share credit, which is why features are permuted in groups.

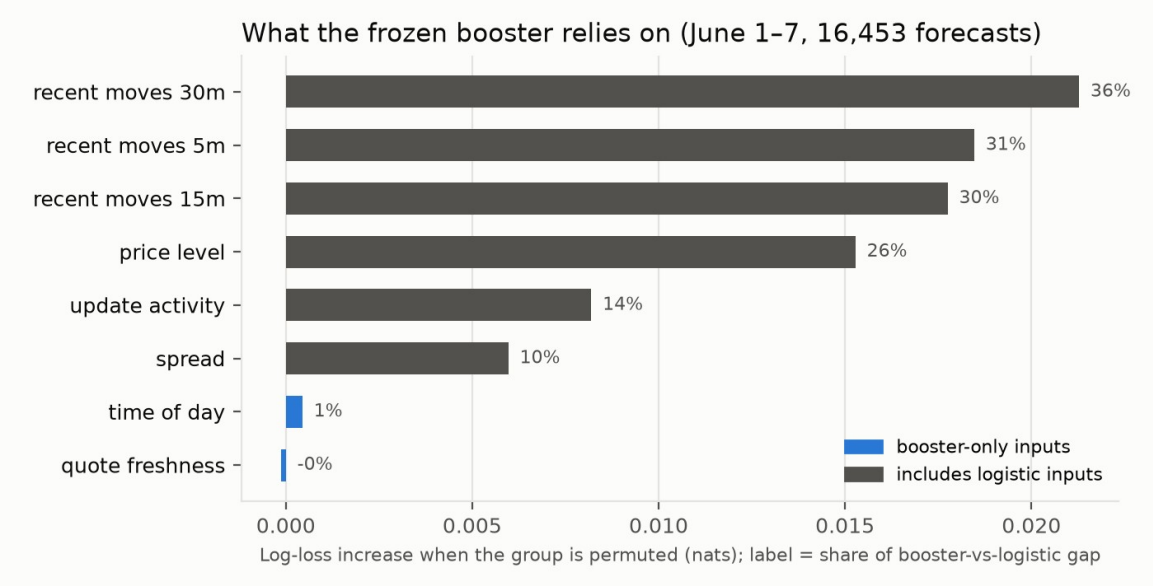


Figure 5. Permutation importance of feature groups for the frozen booster on 1–7 June (16,453 forecasts). Labels give each increase as a share of the booster's log-loss gap over logistic regression.

As a descriptive check, contracts were assigned to categories by a fixed keyword rule on the question text, written before scores were joined. The booster has lower log loss than logistic regression in every observed category in both weeks (Table 7, Figure 6), so gains appear across all observed categories. This does not show how much each category contributes to the aggregate gain. Most categories contain only two to twelve contracts, so these are descriptions, not separate tests.

Table 7. Booster log-loss reduction against activity logistic regression by category (descriptive, post hoc keyword rule).

Block / category	Contracts	Known	Reduction	Contracts won	95% contract CI (nats)
June 1–7 sports	5	5,670	22.1%	5/5	[0.038, 0.103]
June 1–7 finance/crypto	6	3,670	17.5%	5/6	[0.033, 0.101]
June 1–7 politics	5	5,425	11.8%	4/5	[0.001, 0.075]
June 1–7 entertainment	2	1,688	15.9%	2/2	too few
June 8–14 sports	12	11,498	10.3%	9/12	[0.001, 0.096]
June 8–14 finance/crypto	8	7,167	12.1%	6/8	[−0.028, 0.248]
June 8–14 politics	6	6,927	14.3%	5/6	[−0.016, 0.116]
June 8–14 geopolitics	4	6,335	28.8%	4/4	[0.044, 0.241]
June 8–14 tech/science	3	2,680	14.7%	2/3	[−0.038, 0.273]
June 8–14 entertainment	2	611	1.7%	1/2	too few

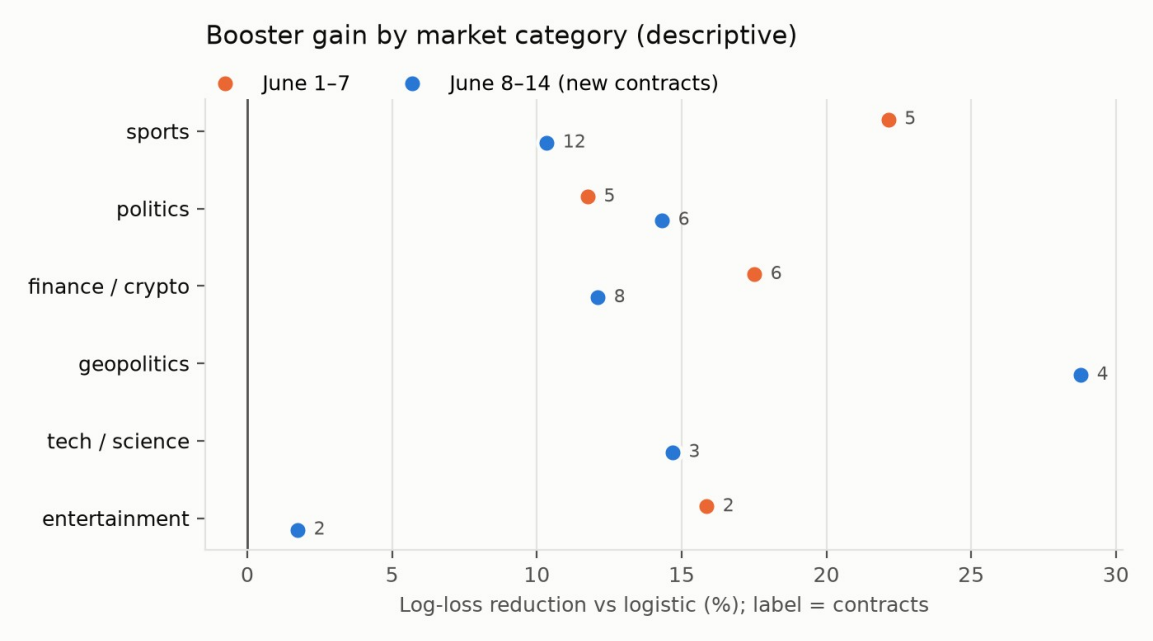


Figure 6. Booster log-loss reduction by market category in each June block. Point labels give the number of contracts.

Post hoc robustness checks. After both June blocks had been consumed, three further checks were run on their stored forecasts (`eventx/tasks/analyze_v7_dependence_coverage_20261028.py`). Nothing was refit or regenerated, and the definitions were fixed in the script before its first run. Two diagnostics added after that run are labelled as such in the output.

Joint dependence. A two-way (pigeonhole) bootstrap resamples contracts and dates independently and weights each row by the product of its contract's and date's multiplicities (Owen, 2007). It assumes contract and date effects combine without a strong interaction, and with seven dates per block the date dimension is estimated imprecisely. The booster's gain over activity logistic regression stays above zero: [0.031, 0.092] nats on 1–7 June and [0.022, 0.126] on 8–14 June, against [0.039, 0.080] and [0.033, 0.119] for contracts alone. Grouping contracts by event gives the same result, because no two contributing contracts share an event. The gain over the legacy booster on 1–7 June does not survive: its two-way interval is [−0.007, 0.092].

Coverage. Of the 96,600 scheduled decisions on 1–7 June, 48.0% fell after the contract's listed end date (current metadata, so approximate), 21.1% in contract-days with no issued forecast and 10.9% within otherwise active contract-days; 32 of the 50 development contracts issued nothing. On 8–14 June the shares were 0.0%, 29.3% and 19.4%, and 5 of 40 contracts issued nothing. Within active contract-days, 64.8% and 72.6% of decisions were issued. The gain over activity logistic regression does not shrink on poorly covered contract-days: in the lowest third

by coverage it is 0.071 and 0.088 nats, against 0.058 and 0.069 in the highest third. Across contracts, the Spearman correlation between coverage and gain is −0.11 (18 contracts) and −0.16 (35). Within the issued forecasts, selection therefore does not appear to favour the booster; nothing can be said about decisions that were never issued.

An online contract rate. None of the fixed baselines learns from a contract's own outcomes. A simple online reference does: at each decision it takes the share of changes among the contract's own earlier outcomes that had already matured (decisions at least 35 minutes earlier), shrunk toward the stored global rate with 20 pseudo-observations, as in the frozen per-contract baseline. It involves no additional offline fitting or tuning: its prior is the stored training-derived global rate, its pseudocount is fixed, and its contract estimate is updated online. Only this one variant was computed. Table 8 compares it with the booster on the two June blocks and on the earlier 13–19 May replication, whose 30 development contracts were also used to train the booster.

Table 8. Post hoc comparison of the frozen booster with an online contract rate built from matured outcomes. Gains are rate minus booster; positive values favour the booster. The May and 1–7 June blocks use contracts from the booster's training cohort.

Block	Booster LL	Online rate LL	LL gain (nats)	95% contract CI	95% contract × date CI	Brier gain [95% CI]	AUC booster / rate	Won: contracts; dates
13–19 May	0.3583	0.3922	0.034 (8.6%)	[0.008, 0.060]	[0.002, 0.071]	0.0170 [0.0075, 0.0275]	0.837 / 0.807	18/30; 7/7
1–7 June	0.2931	0.3143	0.021 (6.7%)	[−0.006, 0.057]	[−0.018, 0.073]	0.0094 [0.0007, 0.0231]	0.835 / 0.790	11/18; 7/7
8–14 June	0.4208	0.4231	0.002 (0.5%)	[−0.028, 0.033]	[−0.040, 0.041]	0.0080 [−0.0016, 0.0178]	0.827 / 0.824	17/35; 3/7

On the 35 new contracts, the online rate reaches almost the booster's log loss and AUC, and both its intervals include zero. On the development contracts, which were also used to train the booster, the booster is ahead: clearly on 13–19 May, where both intervals lie above zero, and on every date of 1–7 June, where both intervals nonetheless include zero. Intervals this wide do not show that the two forecasts are equivalent on the new contracts; they show that the block cannot separate them. The familiar and new cohorts also differ in contracts and dates, so the contrast between them does not isolate an effect of training familiarity.

Outcome history. Grouping decisions by how many of the contract's outcomes had matured shows where the booster gains (`eventx/tasks/analyze_v7_online_rate_replication_20261028.py`). With fewer than 12 matured outcomes, under an hour's worth, it beats the online rate by 0.07–0.14 nats in all three blocks, and four of the six contract-clustered intervals exclude zero. With 12 or more on the new contracts, the difference is about zero. These early decisions are under 2% of forecasts. They account for about 82% of the small aggregate gain on the new contracts, but only 4–7% in the two familiar blocks, where most of the gain comes after history has accumulated. History restarts at each block boundary, so "little history" means few matured outcomes within the analysed block, not necessarily a newly listed contract. Both forecasts are available from a contract's first decision: with no matured outcomes the online rate returns its global prior.

Taken together, these post hoc analyses suggest that quote features are particularly useful when little contract-specific outcome history is available; their incremental value after substantial history accumulates remains uncertain on unseen contracts. The similar warm-period scores are consistent with persistent contract-level activity explaining much of the gain over activity logistic regression, but they do not establish that mechanism. None of this changes a frozen result or pass rule. It does change what those results show: the booster beats the specified fixed baselines, not every simple reference. When information pooled across contracts adds value beyond each contract's own history is a question for new data (Section 9.3).

6.5 A further unopened fortnight: 15–28 June

A third test on 15–28 June uses 21 further never-used contracts, frozen from metadata in a new identifier range before any price was read. An earlier selection with a stricter end-date rule found only 15 contracts. It was superseded before any data was read and is preserved. Both frozen models are scored in a single once-only evaluation under the June 8–14 rule, including the positive contract-clustered lower bound (Table 9). Capture was deferred until after the live window to avoid loading the live host.

Table 9. Matched results, 15–28 June, never-used contracts. Gains are comparator minus booster log loss.

Model vs comparator	Log loss	Reduction	95% contract CI (nats)	Known / missing	Contracts won	Frozen rule
row_hgb7 vs activity logistic	0.3859 / 0.4249	9.2%	[0.022, 0.058]	68,046 / 1,639	17/21	Pass

Model vs comparator	Log loss	Reduction	95% contract CI (nats)	Known / missing	Contracts won	Frozen rule
V03 hgb7 vs its full logistic	0.3883 / 0.4411	12.0%	[0.023, 0.085]	68,700 / 1,336	16/21	Pass

Both frozen rules were met. row_hgb7 had log loss 0.3859 against 0.4249 for activity logistic regression, a 9.2% reduction, with Brier 0.1145 against 0.1241. It had lower log loss on 17 of 21 contracts and on all 14 dates. Its missing-outcome bounds over all 69,685 issued forecasts, [0.029, 0.058] nats, stay above zero. Against the global rate the gain was 0.094 nats [0.062, 0.135]. V03 hgb7 had log loss 0.3883 against 0.4411 for its full logistic regression, a 12.0% reduction, with Brier 0.1160 against 0.1299, and won on 16 of 21 contracts. Two results temper this. First, both boosters had a lower AUC than their logistic comparators (0.744 against 0.783, and 0.747 against 0.779), so on this block the log-loss gain reflects better probability levels rather than better ranking. Second, against the earlier boosted baseline legacy_hgb7 the gain of 0.025 nats has a contract interval of [−0.029, 0.083], which includes zero. The online contract-rate comparison of Section 6.4 was not prespecified for this block; it entered the development data of Section 6.6 instead.

6.6 Confirmatory test: pooled information beyond a contract's own history

Section 6.4 left open whether information pooled across contracts adds value beyond each contract's own outcome history. A new, separately frozen research cycle tested this on new data. Development used only the four already-opened blocks (13–19 May and 1–28 June) and their stored forecasts (`eventx/tasks/develop_v7_history_stack_20261031.py`). Three candidates combining the frozen row_hgb7 forecast, the online contract rate and the number of the contract's matured outcomes were fixed in advance and compared by leave-one-block-out log loss on the two new-contract blocks. The simplest, a logistic regression on the logit of the booster, the logit of the rate and log(1 + matured outcomes) (stack_lin), was selected and refit on all four blocks. Held out, it had lower log loss than both components on every block, for example 0.3750 against 0.4231 for the rate and 0.4208 for the booster on 8–14 June. These development scores were all on opened data.

The protocol, model, code and cohort rule were then hash-locked before any test contract was chosen (`eventx/release/v7/history_stack_20261031/`). The test uses 29 June–12 July, a period no study had read, and contracts from a new identifier range, selected from metadata alone. Of a target of 150, 53 met the frozen eligibility rules and none were added by loosening them. The pass rule required lower log loss than the online rate, a Brier score no

higher, and a contract × date two-way bootstrap lower bound above zero. The window was captured and evaluated once (Table 10).

Table 10. Confirmatory history-stack test, 29 June–12 July, 51 contributing new contracts, 65,458 known outcomes. Gains are comparator minus candidate log loss.

Candidate vs comparator	Log loss	Reduction	95% contract × date CI (nats)	95% contract CI (nats)	AUC	Role
stack vs online rate	0.3348 / 0.3863	13.3%	[0.034, 0.071]	[0.037, 0.066]	0.859 / 0.794	Primary: pass
stack vs row_hgb7	0.3348 / 0.3649	8.3%	[0.011, 0.052]	[0.013, 0.049]	0.859 / 0.799	Secondary
row_hgb7 vs online rate	0.3649 / 0.3863	5.5%	[−0.014, 0.057]	[−0.008, 0.050]	0.799 / 0.794	Secondary
stack vs activity logistic	0.3348 / 0.4255	21.3%	[0.068, 0.116]	[0.072, 0.111]	0.859 / 0.781	Secondary

The frozen rule was met. The stack had log loss 0.3348 against 0.3863 for the online rate, a gain of 0.0515 nats, with Brier 0.1041 against 0.1233. It had lower log loss on 42 of 51 contracts and on all 14 dates. The gain was positive in every history stratum, including decisions with 48 or more matured outcomes (62,716 rows, 0.052 nats, contract interval [0.037, 0.066]), so the pooled forecast kept adding value after a contract's own history had accumulated. The stack also beat the booster alone, while the booster's own gain over the rate, 0.021 nats, had intervals that include zero, as on 8–14 June: each component is roughly as good as the other, but they are not redundant.

Three qualifications apply. First, the frozen capture limit of 32 MiB per hour was exceeded in 134 of 336 hours, mostly busy ones, so no forecasts were issued or scored in those hours; the result describes the remaining 60% of hours. Second, among the 74,197 issued forecasts, 8,739 had no observed endpoint, and the worst-case missing-outcome bounds, [−0.003, 0.083] nats, include zero. Third, as in all historical blocks, cohorts use current metadata and one venue. The test answers the question for this combination and period; it does not show which quote features carry the pooled information.

6.7 Where the gain comes from: segment discovery and replication

A post hoc analysis asked whether the booster's advantage is concentrated in particular markets or times, and whether any such pattern recurs (script in Appendix C). Before any segment score was computed, 30 segments were fixed: market category, UTC hour, price level, spread, recent quote activity, whether the midpoint had moved in the last 30 minutes, the contract's outcome

history and time to its listed end. For each segment, the gain over a comparator was compared with the gain in the rest of the block (the interaction), with contract-clustered intervals. A segment was flagged if the pooled interaction on the three earlier blocks (13–19 May, 1–7 June, 8–14 June) excluded zero and at least two of the three blocks agreed in sign. Flagged segments were then checked on the two later blocks (15–28 June and 29 June–12 July), with a Holm correction. All five blocks had already been opened, so this is descriptive.

Table 11. Booster gain over each comparator by recent quote movement, pooled 15–28 June and 29 June–12 July (133,504 known outcomes, 72 contracts). Gains are comparator minus booster log loss; intervals resample contracts.

Segment	Share of rows	Change rate	vs online rate (nats)	vs activity logistic (nats)
Midpoint moved in last 30 min	16.9%	0.551	+0.088 [0.042, 0.133]	+0.239 [0.192, 0.291]
No move in last 30 min	83.1%	0.109	−0.025 [−0.048, −0.001]	+0.011 [−0.003, 0.025]

Ten segments were flagged and eight replicated (Holm-adjusted p = 0.005 each). They describe two patterns. First, the booster's advantage is concentrated just after a quote move: against the online rate its gain on recently moved rows was positive in all five blocks (0.06 to 0.27 nats), while on quiet rows the rate was as good or slightly better (Table 11). About 80% of the replication-block gain over activity logistic regression came from the 17% of rows with a recent move. Second, the advantage grows with the spread: against the online rate it was +0.091 nats [0.044, 0.149] where the spread exceeded 0.03, and slightly negative, −0.024 nats, at spreads of 0.01 or less. No market category or time of day was flagged, and the price-level and time-to-end flags did not replicate. Added after the first run, the frozen history stack of Section 6.6 kept an edge over the rate in both states on 29 June–12 July: +0.135 nats [0.106, 0.170] after a move and +0.036 [0.022, 0.051] in quiet states, which explains why combining the two sources beat either alone. Because every block had been opened, the two main patterns were then tested on new data (Section 6.8).

6.8 Confirmatory segment test, 13–21 July

The two main patterns of Section 6.7 and the history-stack result were then tested together on new data. Before any contract was chosen, a protocol was hash-locked (`eventx/release/v7/segment_confirm_20261101/`) with two primary hypotheses about the frozen booster's gain over the online rate: that it is larger on rows whose midpoint moved in the previous 30 minutes than on the remaining rows (H1), and larger where the spread

exceeds 0.03 (H2). Each is judged by a one-sided contract-clustered bootstrap p-value for the interaction (5,000 resamples), with a Holm correction over the two. A third hypothesis (H3) re-applies the unchanged history-stack rule of Section 6.6. A routing rule that uses the booster after a move or at a wide spread and the rate otherwise was fixed as a secondary comparison. The test uses 13–21 July, a period no study had read, and 80 contracts from a new identifier range, disjoint from every earlier cohort. Before selection, a dry run on already-consumed July data checked that the evaluation code reproduced the stored forecasts exactly. Two disclosed changes were made before any economic data were read: the capture limit was raised from 32 to 192 MiB per hour, and, after the first selection attempt exceeded its own read limit while reading only identifiers and clocks, a v2 runner raised that limit (Appendix B). All 216 hours were captured, none missing, and the evaluation ran once (Table 12).

Table 12. Confirmatory segment test, 13–21 July, 78 contributing new contracts, 159,283 known outcomes. Gains are online-rate minus candidate log loss (nats); the interaction is the gain in the segment minus the gain on the remaining rows.

Hypothesis	Segment rows (contracts)	Gain in segment	Gain elsewhere	Interaction or gain [95% CI]	Holm p	Result
H1: booster, midpoint moved in last 30 min	28,289 (72)	+0.091	−0.021	0.113 [0.070, 0.162]	0.0004	Confirmed
H2: booster, spread above 0.03	9,331 (40)	+0.084	−0.007	0.091 [0.040, 0.141]	0.0004	Confirmed
H3: stack vs online rate, all rows	159,283 (78)	+0.039	—	0.039 [0.028, 0.051] (contract × date)	—	Pass

Both segment hypotheses were confirmed. On the 17.8% of rows with a recent move the booster beat the rate by 0.091 nats, while on quiet rows the rate was slightly better (−0.021 nats), so the interaction, 0.113 nats, matched the value found in the replication blocks. Where the spread exceeded 0.03 (5.9% of rows) the gain was 0.084 nats against −0.007 elsewhere. The history stack passed again: log loss 0.3339 against 0.3730 for the rate (10.5% lower), Brier 0.1038 against 0.1184, lower log loss on 67 of 78 contracts, and missing-outcome bounds over all 163,818 issued forecasts of [0.028, 0.045] nats, which, unlike in Section 6.6, exclude zero. The stack beat the rate both after moves (+0.128 nats) and in quiet states (+0.020). The routing rule beat the rate by 0.017 nats [0.007, 0.030] but was worse than the stack by 0.022 nats [0.010, 0.035], so a hard switch between the two sources is dominated by their learned combination.

Splitting the same test by market state shows where the forecasting gain is concentrated (Table 13). Active states, in which the midpoint moved in the last 30 minutes or the spread exceeded 0.03, made up 21.4% of rows, and about half of them changed again within 30 minutes. There the booster's log loss was 11.0% below the contract's own rate and the stack's 15.7% below it, 28.6% below activity logistic regression. In the quiet 78.6% of rows the contract's rate was the better single forecaster: the booster was 8.5% worse, while the stack was still 6.7% better. A pooled quote model therefore earns its keep at the moments when a market is moving, which are also the moments that matter most for monitoring, and combining it with the contract's own history keeps the quiet periods from costing accuracy.

Table 13. Log loss by market state, 13–21 July (descriptive breakdown of the test in Table 12; states may overlap). Reductions are relative to the online contract rate.

State	Share of rows	Change rate	Online rate	Booster	Stack	Booster vs rate	Stack vs rate
Active: moved or spread above 0.03	21.4%	0.502	0.7273	0.6473	0.6130	11.0%	15.7%
Moved in last 30 min	17.8%	0.553	0.7641	0.6728	0.6362	11.9%	16.7%
Spread above 0.03	5.9%	0.419	0.6595	0.5753	0.5523	12.8%	16.2%
Quiet: neither	78.6%	0.096	0.2766	0.3002	0.2581	−8.5%	6.7%
All rows	100%	0.183	0.3730	0.3744	0.3339	−0.4%	10.5%

Two qualifications apply. The test covers nine days on one venue, and scores describe issued forecasts: 163,818 of 198,720 scheduled decisions (82.4%) were issued, and 4,535 issued forecasts had no observed endpoint. The segments show when pooled quote information helps; they do not identify which quote features carry it.

6.9 A segment-aware stack, 22 July–8 August

The confirmed segments suggest a direct improvement: let the combination weights depend on market state. A new cycle, again developed only on opened data, fixed three candidates that add the move and spread signals to the stack's inputs: a logistic model with segment interactions, the same with history interactions, and a shallow gradient-boosted combiner (depth 3) on the booster, the rate, history length, activity logistic regression, the move indicator, the spread and the size of the last 30-minute move (`eventx/tasks/develop_v7_segment_stack_20261101.py`). Leave-one-block-out over the six opened blocks selected the boosted combiner (seg_hgb), whose held-out log loss on

the four new-contract blocks was 0.0059 nats below the stack of Section 6.6, above the go threshold of 0.001 nats fixed in advance.

The pmxt archive ends on 10 August, and 22 July–8 August is the calendar period of the consumed V3.1 holdout, which earlier Track B protocols excluded. With no other unused multi-day window left, the user authorized that period as a disclosed deviation (Appendix B). The V3.1, v1 and V4 markets are in the frozen exclusion list, so none can enter the cohort or the capture; no V3.1 file, label or result was read, and the data source, target and models differ. The protocol, seg_hgb and the runner were hash-locked before 54 new contracts were chosen from metadata alone (`eventx/release/v7/segment_stack_20261101/`). The pass rule required lower log loss than the frozen stack, a Brier score no higher and a contract × date lower bound above zero. All 432 hours were captured, one after an automatic retry, and the evaluation ran once (Table 14).

Table 14. Segment-aware stack test, 22 July–8 August, 52 contributing new contracts, 134,341 known outcomes. Gains are comparator minus candidate log loss.

Candidate vs comparator	Log loss	Reduction	95% contract × date CI (nats)	After a move (nats)	No recent move (nats)	Role
seg_hgb vs stack_lin	0.3006 / 0.3083	2.5%	[0.002, 0.014]	+0.023	+0.005	Primary: pass
seg_hgb vs online rate	0.3006 / 0.3497	14.0%	[0.028, 0.076]	+0.186	+0.022	Secondary
stack_lin vs online rate	0.3083 / 0.3497	11.9%	[0.024, 0.064]	+0.162	+0.017	Secondary
row_hgb7 vs online rate	0.3417 / 0.3497	2.3%	[−0.022, 0.045]	+0.135	−0.017	Secondary

The frozen rule was met. seg_hgb had log loss 0.3006 against 0.3083 for the frozen stack, a gain of 0.0076 nats (contract × date interval [0.002, 0.014]), with Brier 0.0926 against 0.0954 and AUC 0.885 against 0.880. It was better on 17 of 18 dates and on 32 of 52 contracts. Its gain over the stack was largest after a move (+0.023 nats) and at wide spreads (+0.026) but positive in quiet states as well. Three further results repeat earlier findings on this new period: the frozen stack beat the rate again (11.9%, interval [0.024, 0.064] nats), a third confirmation; the booster's gain over the rate was again concentrated after moves (interaction 0.152 nats) and at wide spreads (0.158), with one-sided p = 0.0002 for both; and the booster alone was close to the rate overall. Two limits apply. Only 52.5% of scheduled decisions were issued, and over all 140,973 issued forecasts the worst-case missing-outcome bounds of the gain over the stack,

[−0.001, 0.017] nats, include zero. The improvement over the stack is clear on known outcomes but small; most of the value had already been captured by combining the booster with the rate.

6.10 Recent state-of-the-art learners as baselines

The comparators so far were logistic regression, the online rate and the project's own boosters. To check that the gains are not an artefact of weak references, the recent learners of Section 2.6 were run as baselines on the seven opened blocks under a protocol frozen before any baseline was scored (`eventx/release/v7/sota_baselines_20261101/`, with compute-only amendments v2–v4 in Appendix B). Because every block had already been opened, this is a descriptive benchmark, not a confirmatory test, and its outputs were scored exactly once after a label-free check that every file was complete. It has three tiers. Tier A replaces the frozen booster row_hgb7 with TabM, RealMLP, TabICL and default CatBoost, trained on the same 9,212 rows and 18 features and scored on all 506,723 issued rows. Tier B forecasts each contract's outcome series zero-shot from the 288 grids before the maturity lag, the same information the online rate uses: Moirai-MoE on every row, and Sundial and Time-MoE, which are slower, on the nonoverlapping half-hour grid (84,044 rows), where Moirai-MoE and the rate are rescored for a matched comparison. Tier C trains TabM and RealMLP end-to-end on the 18 features, the logit of the online rate and history length, using six blocks, and compares them with seg_hgb on 22 July–8 August (Table 15).

Table 15. Recent baselines on the seven opened blocks (descriptive; scored once). Log losses are means over blocks; blocks better counts blocks in which the baseline's log loss is below the reference's.

Tier and model	Venue	Reference	Log loss (model / reference)	AUC (model / reference)	Blocks better
A: TabM	ICLR 2025	row_hgb7	0.8229 / 0.3627	0.652 / 0.807	0 of 7
A: RealMLP	NeurIPS 2024	row_hgb7	0.4563 / 0.3627	0.718 / 0.807	0 of 7
A: TabICL	ICML 2025	row_hgb7	0.4331 / 0.3627	0.760 / 0.807	0 of 7
A: CatBoost (default)	NeurIPS 2018	row_hgb7	0.4174 / 0.3627	0.769 / 0.807	0 of 7
B: Moirai-MoE, all rows	ICML 2025	Online rate	0.5599 / 0.3703	–	0 of 7
B: Sundial, half-hour grid	ICML 2025	Online rate	0.3526 / 0.3691	0.862 / 0.814	7 of 7
B: Time-MoE, half-hour grid	ICLR 2025	Online rate	0.3908 / 0.3691	0.835 / 0.814	1 of 7

Tier and model	Venue	Reference	Log loss (model / reference)	AUC (model / reference)	Blocks better
B: Moirai-MoE, half-hour grid	ICML 2025	Online rate	0.5517 / 0.3691	0.797 / 0.814	0 of 7
C: TabM end-to-end	ICLR 2025	seg_hgb	0.3411 / 0.3006	0.849 / 0.885	22 Jul–8 Aug only
C: RealMLP end-to-end	NeurIPS 2024	seg_hgb	0.3241 / 0.3006	0.869 / 0.885	22 Jul–8 Aug only

No tabular baseline beat row_hgb7 on any block. Each was worse in log loss on all seven, with contract-clustered intervals below zero in every case, and its AUC was lower too, so the gap is not only one of calibration. CatBoost was the closest (0.040 to 0.071 nats worse) and TabM the furthest. In Tier C both end-to-end networks were clearly worse than seg_hgb: RealMLP by 0.0234 nats (interval [−0.040, −0.011]) and TabM by 0.0405 ([−0.054, −0.029]), and both were also worse than the plain stack. The one competitive result came from the time-series foundation models. Sundial, used zero-shot on the contract's own outcome series, had lower log loss than the online rate on all seven blocks (gains 0.004 to 0.023 nats, intervals above zero on four) and higher AUC (by 0.030 to 0.067) and lower Brier score on every block. On the three blocks where the frozen stack was available, though, the stack was better still (for example 0.3068 against 0.3270 on 22 July–8 August). Time-MoE beat the rate on one block and Moirai-MoE on none. Time-MoE ranked rows better than the rate (AUC 0.835 against 0.814) but its probabilities were poorly calibrated, while Moirai-MoE was worse on both counts.

Three limits apply. First, Tier A used each package's default internal validation split, which is random rather than chronological, and no baseline was tuned; matched tuning on purged forward folds could narrow the gaps, so these results show that published defaults do not beat the frozen booster, not that the booster is better than every tuned alternative. Second, Tier C trained the networks on a 100,000-row seeded subsample because of the host's compute limits, whereas seg_hgb used all rows, which favours seg_hgb. Third, the Tier B models forecast real-valued series; scoring them here tests one adaptation (a binary series with mean-imputed gaps, the step-7 forecast clipped to [0.001, 0.999]), not these models in general. Within those limits, recency did not supply a stronger reference: the boosted models and the stacks remained the best forecasters, and the only recent model that helped, Sundial, did so as a forecaster of the contract's own history, the part that the stack already captures through the online rate.

6.11 A development study: more training data and change-history features

A last development cycle asked whether the segment-aware stack could be improved, using only the seven opened blocks. Its code was committed before any score (`develop_v8_hawkes_posterior_20261102.py`, Appendix C). It examined three ideas. First, seg_hgb takes the frozen booster, trained on 9,212 rows, as an input; a single booster trained end-to-end on the frozen features, the online rate and history length can learn from all 506,723 issued rows of the opened blocks. Second, the frozen features describe recent quote changes only through 5- and 30-minute differences. Five derived inputs summarise each contract's recent change history: the time since its last 5-minute change (log minutes, capped at one day), exponentially decayed counts of changes with half-lives of 30 minutes, 2 hours and 12 hours, and the number of changes in the previous 24 hours. These are decayed change counts and recency features in the spirit of self-exciting point processes (Hawkes, 1971), not a fitted Hawkes model. Two market-wide inputs add the share of the block's other contracts that moved in the last 30 minutes and the number quoting. Third, a per-contract offset is added to a model's logit and updated by a one-step Laplace approximation each time one of the contract's outcomes matures. This is an approximate online logistic correction of the kind used in dynamic generalised linear models (West, Harrison and Migon, 1985; McCormick et al, 2012); its prior variance and drift were fixed in advance, not fitted. Every candidate used fixed hyperparameters and was scored by leaving one block out and training on the other six. The score is the mean held-out log loss over the five new-contract blocks (Table 16).

Table 16. Development study (opened blocks only): leave-one-block-out log loss, mean over the five new-contract blocks. Gains are seg_ref minus candidate log loss.

Candidate	Inputs	Mean log loss	Gain vs seg_ref (nats)	Blocks better than seg_ref
Online rate	Contract's matured outcomes	0.3771	−0.0489	–
row_hgb7 (frozen)	18 quote features	0.3775	−0.0494	–
seg_ref	seg_hgb design, refit per fold	0.3282	reference	–
e2e	18 features, rate, history	0.3270	+0.0011	3 of 5
e2e_hx	e2e + change history	0.3150	+0.0132	5 of 5
e2e_hx_mkt	e2e_hx + market activity	0.3159	+0.0122	5 of 5
pp_cp	e2e_hx_mkt + online contract offset	0.3229	+0.0052	4 of 5
pp_cp_seg	seg_ref + online contract offset	0.3234	+0.0047	4 of 5

More training data alone (e2e) roughly matched seg_ref. The change-history inputs gave the clear gain: e2e_hx had the lowest score, 0.3150, 4.0% below seg_ref, and lower held-out log loss on all seven folds, with contract-clustered intervals above zero in each (gains 0.0075 to 0.0173 nats). On 22 July–8 August, the only fold trained solely on earlier blocks, the gain was 0.0124 nats [0.0066, 0.0186]. Market-wide activity added nothing on average. The online contract offset improved seg_ref by 0.0047 nats but made its own base model, e2e_hx_mkt, worse by 0.0070 nats; once the change history is in the model, the per-contract offset mostly adds noise. This is retrospective development evidence. In six of the seven folds the training blocks include dates later than the held-out block, the forward fold had already been opened by the test of Section 6.9, and the ranking was not fixed in advance as a test. e2e_hx is therefore the candidate with the lowest development score, to be tested on new data (Section 9.3), not a confirmed improvement.

7 Prospective Live Test

7.1 Design

Historical replay cannot rule out subtle advantages from hindsight, such as metadata that is only known today or receive clocks that are reconstructed. A prospective test removes both. The live runtime is split into a **source** process and an **emitter** process:

- the **source** subscribes to the Polymarket WebSocket for the cohort and polls metadata. It writes every message to an append-only journal in which each record carries the SHA-256 hash of the previous one;
- the **emitter** reads that journal and, at each five-minute grid time, issues a forecast from the frozen V03 model or records an abstention with machine-readable reasons. Each forecast is committed at least 250 ms before its grid time.

An evaluator scores the forecasts once, after the window closes, against endpoints taken from the same journal. The cohort is 30 contracts, chosen by metadata only. They are active, binary, order-book enabled, at least seven days old, resolve no earlier than 10 November, have at least $10k of 24-hour volume (used as a floor only), are taken in salted-hash order with one contract per event, and are disjoint from every earlier cohort.

7.2 Engineering problems and their solutions

Each problem below caused a rehearsal or activation to be stopped and archived, never patched in place.

- **Unreliable host clock.** The first runtime version failed its readiness rehearsal with 0 of 16 decisions ready. The laptop's built-in time client returned empty server packets while the clock was about 287 ms fast, and the operating system then stepped the clock back by 110 ms, which reset the transport warmup. The new version uses its own SNTP client (Mills et al, 2010). It keeps the raw packets and recomputes offsets from them. An estimate is accepted only if at least two servers agree, at least one round trip is short, and the residual uncertainty is at most 50 ms. Corrected time is then carried forward on the monotonic clock, so operating-system clock steps are logged but cannot move decision times.
- **Memory and latency growth.** The first activation kept the whole journal in memory (about 317 MB per hour) and rescanned it at every decision. A bounded, incremental input store keeps three hours of history, more than every rule needs. It matched the

unbounded version in 510 of 510 decision and endpoint comparisons, and cut decision time from 370 ms to 11 ms per grid.

- **A thread race in the hash chain.** In the second activation, a clock-probe thread wrote to the journal at the same time as the main loop, which forked the hash chain. The emitter's chain check detected this and stopped. Now only the event loop writes the journal, and a regression test enforces it.

After these fixes, a fresh 60-minute readiness rehearsal passed. 156 of 180 post-warmup decisions were ready, across 27 of 30 contracts; 58 of 61 clock estimates were valid; one OS clock step was absorbed; and there were 0 chain breaks, stamp errors or replay mismatches. The test was activated at 17:53:53 UTC on 26 October, with grids from 27 October 00:30 to 29 October 23:25 UTC (24,840 scheduled decisions). The window was shortened from seven days to three, before any prospective decision, so that it would finish before this report's deadline.

7.3 Operation

The test has known operating losses, all recorded rather than repaired:

- **First grid.** All 30 decisions abstained, because the emitter's pre-loaded inputs were handed over without one final refresh (0.12% of the schedule).
- **Host stall.** A concurrent analysis job pushed the already memory-constrained laptop into heavy swapping, so four grids missed their emission deadlines and roughly an hour of warmup abstentions followed. Heavy jobs were then deferred until the window closed.
- **Sleep and reboot.** On 27 October at 04:42 UTC the laptop, running on battery with its lid closed, entered clamshell sleep, which `caffeinate` cannot prevent. It then rebooted at 05:16 UTC. The source and emitter did not survive. The frozen protocol forbids restarting the source into the same run, so every later grid in that run counts as abstained or missed. At the 05:19 UTC status check the ledger held 725 ready decisions, 715 abstentions and 300 misses. The run could not be resumed, and its frozen evaluator cannot score it, because the source never wrote its closing capture manifest. With the user's authorisation, a separately versioned continuation (**V02b**) was activated at 05:47 UTC before any V02 outcome was read. V02b reuses every frozen V02 file, the V03 model and the 30-contract cohort unchanged. Only the run directory and the

window differ: grids from 27 October 06:35 to 29 October 23:25 UTC, 22,650 scheduled decisions. V02b is the live result reported in Section 7.5.

- **Second sleep (V02b).** From 07:40 to 08:26 UTC on 27 October the lid was closed again. The laptop was on mains power, but sleep had not yet been disabled system-wide, so it slept. The V02b processes survived, but nine grids (270 decisions) passed their emission deadlines and count as missed. Sleep was then disabled at system level (`pmset disablesleep 1`).
- **Venue disconnects.** Outside the stall and sleep episodes, the venue WebSocket disconnected about every three to six hours: three times in about 9.5 hours of V02, and three times in about 17 hours of V02b up to 28 October 01:40 UTC. Each reconnect costs 30 minutes of abstentions under the frozen warmup rule. Bursts of disconnects coincided with the host stall and the sleeps.

These losses reduce coverage but cannot leak information, since every failure leads to a recorded abstention or miss and nothing is backfilled.

7.4 Lessons for operating a prospective forecaster

The live track produced engineering evidence of its own. Every incident below ended in an abstention rather than a confident wrong forecast, and every one is preserved. The common thread is that a forecaster that abstains honestly can be measured, whereas one that silently fills gaps cannot.

Table 17. Operational incidents and lessons.

Incident	Design response	Lesson
Host clock: empty SNTP packets, −110 ms OS step	Packets rejected; warmup reset; readiness gate failed	Fix the clock, not the gate; own validated SNTP client
Memory growth (317 MB/h)	Stopped before first grid; 0 forecasts	Load-test at full horizon before activation
Thread race in hash chain	Chain check stopped the emitter	Single-writer journals, enforced by test
DNS outage during capture	Outcome-blind refetch to a separate directory	Retry rules need an explicit outage clause
First-grid handoff	30 abstentions (0.12%)	Rehearse the activation handoff end to end
Swap stall from a concurrent job	4 grids missed; warmup abstentions	Separate production and research hosts
Lid-closed sleep and reboot	Run ended; no backfill	Always-on host: AC power, sleep disabled, service restart

Incident	Design response	Lesson
Second lid-closed sleep (V02b, on AC)	9 grids (270 decisions) missed; processes survived	Disable sleep system-wide before activation, not per process

7.5 Results

The live test has no pass rule. Its registered primary comparison is matched log loss of V03 hgb7 against its full logistic regression, with the global rate as a second comparator, and scores are reported only if at least 100 outcomes are known, with at least 20 of each class. The evaluation ran once, at 01:57 UTC on 30 October, and is reported whatever the sign (Table 18).

Table 18. Live V02b test: coverage and matched scores. Gains are comparator minus V03 hgb7 log loss.

Quantity	Value
Scheduled decisions	22,650
Issued on time (coverage)	15,485 (68.4%)
Abstained, missed or uncommitted	7,165
Known / unknown outcomes (positives)	12,943 / 2,542 (2,709)
Reporting floor met	Yes (27 contracts)
Log loss: V03 hgb7 / full logistic / global rate	0.4470 / 0.5368 / 0.5163
Brier: V03 hgb7 / full logistic / global rate	0.1403 / 0.1547 / 0.1665
AUC: V03 hgb7 / full logistic	0.745 / 0.726
Relative log-loss reduction vs full logistic	16.7%
95% contract-clustered interval of the gain (nats)	[0.049, 0.136]
Missing-outcome bounds over all issued forecasts (nats)	[−0.011, 0.150]

The reporting floor was met. 27 of the 30 contracts contributed known outcomes. On the 12,943 known outcomes, V03 hgb7 had log loss 0.4470 against 0.5368 for its full logistic regression (16.7% lower; contract-clustered interval [0.049, 0.136] nats) and 0.5163 for the global rate (interval [0.034, 0.112] nats). Its Brier score and AUC were also the best of the three. The full logistic model scored worse than the global rate in log loss but better in Brier and AUC, which suggests it was miscalibrated on live data. Two limits apply. First, 2,542 issued forecasts (16.4%) have no known outcome, and over all issued forecasts the missing-outcome bounds of the gain are [−0.011, 0.150] nats against the full logistic and [−0.073, 0.102] nats against the global rate, so the known outcomes alone do not fix the sign. Second, the test covers three days, one venue and 68.4% coverage, and has no pass rule. It is a descriptive prospective result, consistent in direction with the historical blocks.

8 Discussion

8.1 What the evidence supports

The answers to RQ1 and RQ2 differ sharply. Simple news counts did not improve a market-history forecaster under strict availability rules, while a microstructure target was forecast better than the specified baselines on all three evaluated unopened June blocks, and in the same direction on the known outcomes of the live test. This is consistent with a market in which public news is either quickly priced or too weakly linked to individual contracts to be detected by counts, whereas quote activity is persistent and well summarised by recent quote changes, spread and price level. Why the booster wins is not settled. Jointly permuting the inputs unavailable to logistic regression removed about a fifth of its advantage, but as a sensitivity analysis of a fitted model this does not attribute the advantage to either those inputs or nonlinearity. A post hoc online contract rate, which only tracks each contract's own realised change rate, came close to the booster on the new contracts (Table 8). This is consistent with much of the advantage over activity logistic regression coming from persistent contract-level activity, which a fixed nine-input logistic model does not capture, but it does not establish that explanation. The confirmatory test of Section 6.6 then showed that the two sources are complementary: combining the booster with the online rate beat the rate alone by 13.3% on new contracts and dates, including after many outcomes had matured. Pooled quote information therefore adds value beyond a contract's own history in this setting. A post hoc segment analysis locates most of that value just after quote moves and in wide-spread states, a pattern that recurred on two later blocks, while no market category or time of day stood out (Section 6.7). A frozen test on 78 further new contracts (13–21 July) confirmed both patterns and the stack's gain over the rate (Section 6.8), and the pattern recurred on 22 July–8 August. Using it, a segment-aware stack improved on the plain stack by 2.5% in a once-only test (Section 6.9). Recent tabular and time-series learners did not provide a stronger reference when run with their default settings (Section 6.10), and a development study suggests that a contract's recent change history carries further signal (Section 6.11). Segmenting by market state therefore turned an apparent tie between a pooled model and a contract's own history into a clear, repeatedly confirmed gain in active states and a usable improvement overall.

8.2 Threats to validity

- **Dependence.** Forecasts from one contract are strongly correlated. This is why the headline intervals resample whole contracts, and why the effective sample size is closer

to the number of contracts (18–35) than to the number of rows. Contracts also share calendar dates and therefore common information shocks. A post hoc two-way bootstrap over contracts and dates widens the intervals but leaves both June gains over activity logistic regression above zero (Section 6.4). With only seven dates per block, the date dimension is itself estimated imprecisely.

- **Comparator strength.** The primary comparator uses nine of the booster's 18 inputs (Section 5.4), and no full-feature logistic regression, smooth nonlinear model or comparably tuned alternative ensemble was fitted for the June blocks. A post hoc online contract rate came within 0.5% of the booster's log loss on 8–14 June, although the booster was clearly ahead of it on 13–19 May (Table 8). The June results show improvement over the specified baselines, not over the strongest reasonable baseline, and they do not attribute the gain to nonlinearity. The 29 June–12 July test used the online rate as its primary comparator, but its development candidates were few and simple, so a stronger history-only model might narrow the gap.
- **Coverage and selection.** Forecasts were issued for 19,304 of 96,600 scheduled decisions in 1–7 June (20.0%) and 39,656 of 77,280 in 8–14 June (51.3%). On 1–7 June, 48.0% of scheduled decisions fell after the contract's listed end, and on both blocks most of the rest fell in contract-days with no valid quote (Section 6.4), so the shortfall mainly reflects inactive or expired contracts and the frozen quote-validity gates rather than feed failures. Within the issued forecasts the gain does not shrink on poorly covered contract-days, but the scores still describe the issued, active part of the market. The missing-outcome bounds cover unknown endpoints among issued forecasts, not decisions that were never issued. In the 29 June–12 July test, a frozen per-hour capture limit removed 134 of 336 hours, mostly busy ones, so that test describes quieter hours more than busy ones.
- **Temporal breadth.** The historical evidence spans 1 June–12 July in four blocks. Chronological transfer on familiar contracts and transfer to new contracts have been tested; transfer to new event families, sustained prospective issuance and transfer to another venue have not.
- **Cohort and venue.** Cohorts are convenience samples that require quotes to be available, and all Track B evidence comes from one venue and one era (2026). The

new-contract test reduces, but does not remove, the risk that results depend on contract identity.

- **Metadata timing.** Historical cohorts use current metadata, so a contract's present description may differ from what was known in June. Only the live test is free of this.
- **Conditioning on observed endpoints.** Scores use only forecasts with observed endpoints. The worst-case bounds show that missing endpoints cannot reverse the June 8–14 conclusion, but in June 1–7 the lower bound falls just below zero.
- **Protocol deviations.** The DNS refetch and the V3.1 prevalence exposure are real departures from the intended protocols. They were outcome-blind or happened before any metric was computed, but they are disclosed rather than argued away.
- **Causal language.** None of the results is causal. Track A estimates predictive value conditional on observable news; information that surfaces first on unobserved channels (private chats, livestreams) remains a confounder.

9 Conclusions

9.1 Summary

EventX built a leakage-safe evaluation framework for prediction-market forecasting and applied it to two questions. In a frozen study of 4,651 markets, simple news-count augmentation failed the specified development promotion gates. A frozen gradient-boosted quote-change classifier improved matched historical scores against the specified baselines on two unopened June blocks, including a cohort of 35 never-used contracts: log loss fell by 16.8% and 14.9% against activity logistic regression, with contract-clustered intervals above zero. Post hoc, a contract's own online outcome rate came within 0.5% of the booster's log loss on the new contracts, so the gain is over the specified fixed baselines rather than over every simple reference; quote features appeared most useful when little contract-specific outcome history was available. A prospective runtime with validated clock correction, bounded inputs and tamper-evident journals was built, and a three-day live test was run. A third unopened block (15–28 June, 21 new contracts) met the frozen rule, with 9.2% lower log loss and an interval of [0.022, 0.058] nats. A confirmatory test on 51 further new contracts (29 June–12 July) then showed that a frozen logistic stack of the booster and the contract's online rate beat the rate alone by 13.3% in log loss (contract × date interval [0.034, 0.071] nats), so pooled quote information adds value beyond each contract's own history. A second frozen test on 78 new contracts (13–21 July) repeated this (10.5%, interval [0.028, 0.051] nats) and confirmed that the booster's gain over the rate is concentrated after recent quote moves and at wide spreads. A stack that uses those two signals then beat the plain stack on 52 further new contracts (22 July–8 August; 0.3006 against 0.3083, interval [0.002, 0.014] nats). Live, 68.4% of scheduled forecasts were issued on time, and on 12,943 known outcomes V03 hgb7's log loss was 16.7% below its logistic baseline, a descriptive result whose missing-outcome bounds include zero. The richer-news V4 test, run once after a disclosed revision of its data audit, did not improve on market history, although all its models transferred poorly to that period.

9.2 Limitations

- The quote-change target measures activity, not direction, size or profit. A good score does not imply a trading edge.
- Track B uses one venue and small numbers of contributing contracts, and historical cohorts rely on current metadata.

- The primary comparator is weaker than a matched-input baseline, so the June gains are not attributed to nonlinearity. A post hoc online contract rate scored close to the booster on the new contracts. Two later confirmatory tests showed that their combination beats the rate: on 51 contracts with 60% of hours captured and missing-outcome bounds that include zero, and on 78 contracts over nine fully captured days with bounds that exclude zero. The segment-aware stack's gain over the stack was tested in the V3.1 holdout calendar period by authorized deviation, and its missing-outcome bounds include zero. Scores describe issued forecasts, which covered 20.0% and 51.3% of scheduled decisions in the two June blocks.
- Track A tested count-based news representations under conservative timestamps, and V4 tested richer news features (relevance, novelty, publisher diffusion) on a period to which its market-history base model transferred poorly. The negative results apply to those representations; it does not show that public news carries no information. Contract–event links and first-availability times were not independently verified, and richer text models and social-media signals were not evaluated within the frozen protocols.
- The live test ran on a laptop whose memory pressure, sleeps and network drops reduced coverage. A disciplined server would give a cleaner prospective estimate.
- The news bound, permutation importance and category breakdown are post hoc descriptive analyses of already-opened data; none was used to change a frozen model.

9.3 Recommendations for further work

1. Extend the history-stack result of Section 6.6. Capture busy hours in full, add a recency-weighted rate prespecified on new development data as a stronger history-only comparator, and compare logistic, GAM and boosting combinations with the rates alone on more contracts, longer periods and by outcome history. An equivalence test with a margin justified in advance could show that a warm-period difference is smaller than a practically relevant size, though not that it is zero, and an inconclusive result remains possible (Lakens, 2017).
2. Test the segment-aware stack of Section 6.9 prospectively on live data, where no archive limit or reused calendar period applies, and test whether news features help when restricted to the active states in which pooled quote information is most useful.

3. Freeze e2e_hx, the end-to-end booster with change-history inputs that had the lowest development score in Section 6.11, and test it once against seg_hgb on new contracts and dates under the same pass rule. All historical archive windows are now used, so this test must be prospective.
4. Test whether the forecasts help allocate a fixed monitoring budget, against uniform, round-robin, recent-activity and online-rate rankings at equal total cost. The design must not assume fresh quotes it would have to pay for: either allocate downstream analysis given an existing quote feed, or simulate polling with stale observations and charged refreshes. “Captured changes” must be defined in advance (endpoint differences, distinct movements or detection latency) so that overlapping 30-minute windows do not count one movement repeatedly. A gain here would show monitoring utility, not trading profit.
5. Compare learners and inputs factorially: logistic regression, a spline or GAM model and gradient boosting, each on the same nine activity inputs and on the same full input set, with regularisation and tuning budgets matched and selected on purged forward development folds. Add an adaptive activity-rate baseline that uses only outcomes already observed at each decision. This separates the contribution of nonlinearity from that of the extra inputs, which permutation importance cannot do.
6. Test calibration on a dedicated development calibration split, scoring log loss, Brier and reliability, since calibration can improve probability quality without changing AUC.
7. Run the live runtime on a dedicated host with a disciplined clock and enough memory, for at least seven days, on a larger metadata-selected cohort.
8. Extend the target to signed and size-graded quote moves, which are closer to economically meaningful questions, keeping the same once-only protocol.
9. Re-test news value only after contract–event links are independently reviewed and release and first-availability times are evidenced, comparing market-only, news-volume, context-only and content-based models on identical rows, with the quote-change model as the market-only baseline.
10. Widen the independent evidence: more event families, longer periods and different liquidity and expiry regimes, with the number of contracts set by cluster-level precision planning on development data rather than by row counts. As a preliminary guide only, scaling the 8–14 June contract-clustered interval against the online rate suggests several

hundred contracts (roughly 660 for 80% power at 0.01 nats). That figure ignores date dependence, the length of the test period and the loss of inactive contracts, all of which must be planned jointly before a study is frozen.

11. Replicate on Kalshi, whose historical intraday data has a different coverage pattern, to test venue transfer.

References

Arrow, K.J., Forsythe, R., Gorham, M., Hahn, R., Hanson, R., Ledyard, J.O., Levmore, S., Litan, R., Milgrom, P., Nelson, F.D., Neumann, G.R., Ottaviani, M., Schelling, T.C., Shiller, R.J., Smith, V.L., Snowberg, E., Sunstein, C.R., Tetlock, P.C., Tetlock, P.E., Varian, H.R., Wolfers, J. and Zitzewitz, E. (2008). The Promise of Prediction Markets. *Science*, Vol. 320, No. 5878, pp. 877 – 878.

Brier, G.W. (1950). Verification of Forecasts Expressed in Terms of Probability. *Monthly Weather Review*, Vol. 78, No. 1, pp. 1 – 3.

Cameron, A.C., Gelbach, J.B. and Miller, D.L. (2008). Bootstrap-Based Improvements for Inference with Clustered Errors. *Review of Economics and Statistics*, Vol. 90, No. 3, pp. 414 – 427.

Croxson, K. and Reade, J.J. (2014). Information and Efficiency: Goal Arrival in Soccer Betting. *The Economic Journal*, Vol. 124, No. 575, pp. 62 – 91.

Diercks, A.M., Katz, J.D. and Wright, J.H. (2026). Kalshi and the Rise of Macro Markets. NBER Working Paper No. 34702, National Bureau of Economic Research, Cambridge, Massachusetts.

Engle, R.F. and Russell, J.R. (1998). Autoregressive Conditional Duration: A New Model for Irregularly Spaced Transaction Data. *Econometrica*, Vol. 66, No. 5, pp. 1127 – 1162.

Erickson, N., Purucker, L., Tschalzev, A., Holzmüller, D., Desai, P.M., Salinas, D. and Hutter, F. (2025). TabArena: A Living Benchmark for Machine Learning on Tabular Data. In *Advances in Neural Information Processing Systems 38* (Datasets and Benchmarks Track, San Diego, California, December 2025).

Friedman, J.H. (2001). Greedy Function Approximation: A Gradient Boosting Machine. *Annals of Statistics*, Vol. 29, No. 5, pp. 1189 – 1232.

Gneiting, T. and Raftery, A.E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, Vol. 102, No. 477, pp. 359 – 378.

Gorishniy, Y., Kotelnikov, A. and Babenko, A. (2025). TabM: Advancing Tabular Deep Learning with Parameter-Efficient Ensembling. In *Proceedings of the Thirteenth*

International Conference on Learning Representations (ICLR 2025, Singapore, April 24 – 28, 2025).

Grinsztajn, L., Oyallon, E. and Varoquaux, G. (2022). Why Do Tree-Based Models Still Outperform Deep Learning on Typical Tabular Data? *Advances in Neural Information Processing Systems 35* (Datasets and Benchmarks Track), pp. 507 – 520.

Halawi, D., Zhang, F., Yueh-Han, C. and Steinhardt, J. (2024). Approaching Human-Level Forecasting with Language Models. In *Advances in Neural Information Processing Systems 37* (Vancouver, Canada, December 10 – 15, 2024).

Hasbrouck, J. (2007). *Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading.* Oxford University Press, New York, 2007.

Hawkes, A.G. (1971). Spectra of Some Self-Exciting and Mutually Exciting Point Processes. *Biometrika*, Vol. 58, No. 1, pp. 83 – 90.

Holzmüller, D., Grinsztajn, L. and Steinwart, I. (2024). Better by Default: Strong Pre-Tuned MLPs and Boosted Trees on Tabular Data. In *Advances in Neural Information Processing Systems 37* (Vancouver, Canada, December 10 – 15, 2024).

Kapoor, S. and Narayanan, A. (2023). Leakage and the Reproducibility Crisis in Machine-Learning-Based Science. *Patterns*, Vol. 4, No. 9, Article 100804.

Karger, E., Bastani, H., Yueh-Han, C., Jacobs, Z., Halawi, D., Zhang, F. and Tetlock, P.E. (2025). ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities. In *Proceedings of the Thirteenth International Conference on Learning Representations* (ICLR 2025, Singapore, April 24 – 28, 2025).

Kaufman, S., Rosset, S., Perlich, C. and Stitelman, O. (2012). Leakage in Data Mining: Formulation, Detection, and Avoidance. *ACM Transactions on Knowledge Discovery from Data*, Vol. 6, No. 4, Article 15.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.-Y. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30* (Long Beach, California, December 4 – 9, 2017), pp. 3146 – 3154.

Kolm, P.N., Turiel, J. and Westray, N. (2023). Deep Order Flow Imbalance: Extracting Alpha at Multiple Horizons from the Limit Order Book. *Mathematical Finance*, Vol. 33, No. 4, pp. 1044 – 1081.

Künsch, H.R. (1989). The Jackknife and the Bootstrap for General Stationary Observations. *Annals of Statistics*, Vol. 17, No. 3, pp. 1217 – 1241.

Lakens, D. (2017). Equivalence Tests: A Practical Primer for t Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science*, Vol. 8, No. 4, pp. 355 – 362.

Liu, X., Liu, J., Woo, G., Aksu, T., Liang, Y., Zimmermann, R., Liu, C., Li, J., Savarese, S., Xiong, C. and Sahoo, D. (2025). Moirai-MoE: Empowering Time Series Foundation Models with Sparse Mixture of Experts. In *Proceedings of the 42nd International Conference on Machine Learning* (Vancouver, Canada, July 13 – 19, 2025). PMLR, Vol. 267, pp. 38940 – 38962.

Liu, Y., Qin, G., Shi, Z., Chen, Z., Yang, C., Huang, X., Wang, J. and Long, M. (2025b). Sundial: A Family of Highly Capable Time Series Foundation Models. In *Proceedings of the 42nd International Conference on Machine Learning* (Vancouver, Canada, July 13 – 19, 2025). PMLR, Vol. 267, pp. 39295 – 39317.

López de Prado, M. (2018). *Advances in Financial Machine Learning.* John Wiley & Sons, Hoboken, New Jersey, 2018.

McCormick, T.H., Raftery, A.E., Madigan, D. and Burd, R.S. (2012). Dynamic Logistic Regression and Dynamic Model Averaging for Binary Classification. *Biometrics*, Vol. 68, No. 1, pp. 23 – 30.

Mills, D., Martin, J. (Ed.), Burbank, J. and Kasch, W. (2010). Network Time Protocol Version 4: Protocol and Algorithms Specification. RFC 5905, Internet Engineering Task Force, June 2010.

Niculescu-Mizil, A. and Caruana, R. (2005). Predicting Good Probabilities with Supervised Learning. In *Proceedings of the 22nd International Conference on Machine Learning* (Bonn, Germany, August 7 – 11, 2005). ACM, New York, 2005, pp. 625 – 632.

Ntakaris, A., Magris, M., Kanniainen, J., Gabbouj, M. and Iosifidis, A. (2018). Benchmark Dataset for Mid-Price Forecasting of Limit Order Book Data with Machine Learning Methods. *Journal of Forecasting*, Vol. 37, No. 8, pp. 852 – 866.

Owen, A.B. (2007). The Pigeonhole Bootstrap. *The Annals of Applied Statistics*, Vol. 1, No. 2, pp. 386 – 411.

O'Hara, M. (1995). *Market Microstructure Theory.* Blackwell Publishers, Cambridge, Massachusetts, 1995.

Page, L. and Clemen, R.T. (2013). Do Prediction Markets Produce Well-Calibrated Probability Forecasts? *The Economic Journal*, Vol. 123, No. 568, pp. 491 – 513.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M. and Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, Vol. 12, pp. 2825 – 2830.

Polymarket. (2026). Polymarket CLOB and Gamma API Documentation. Online documentation, accessed October 2026.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V. and Gulin, A. (2018). CatBoost: Unbiased Boosting with Categorical Features. In *Advances in Neural Information Processing Systems 31* (Montréal, Canada, December 3 – 8, 2018), pp. 6638 – 6648.

Qu, J., Holzmüller, D., Varoquaux, G. and Le Morvan, M. (2025). TabICL: A Tabular Foundation Model for In-Context Learning on Large Data. In *Proceedings of the 42nd International Conference on Machine Learning* (Vancouver, Canada, July 13 – 19, 2025). PMLR, Vol. 267, pp. 50817 – 50847.

Shi, X., Wang, S., Nie, Y., Li, D., Ye, Z., Wen, Q. and Jin, M. (2025). Time-MoE: Billion-Scale Time Series Foundation Models with Mixture of Experts. In *Proceedings of the Thirteenth International Conference on Learning Representations* (ICLR 2025, Singapore, April 24 – 28, 2025).

Sirignano, J. and Cont, R. (2019). Universal Features of Price Formation in Financial Markets: Perspectives from Deep Learning. *Quantitative Finance*, Vol. 19, No. 9, pp. 1449 – 1459.

Snowberg, E., Wolfers, J. and Zitzewitz, E. (2007). Partisan Impacts on the Economy: Evidence from Prediction Markets and Close Elections. *Quarterly Journal of Economics*, Vol. 122, No. 2, pp. 807 – 829.

Tetlock, P.C. (2007). Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *Journal of Finance*, Vol. 62, No. 3, pp. 1139 – 1168.

West, M., Harrison, P.J. and Migon, H.S. (1985). Dynamic Generalized Linear Models and Bayesian Forecasting. *Journal of the American Statistical Association*, Vol. 80, No. 389, pp. 73 – 83.

Wolfers, J. and Zitzewitz, E. (2004). Prediction Markets. *Journal of Economic Perspectives*, Vol. 18, No. 2, pp. 107 – 126.

Zhang, Z., Zohren, S. and Roberts, S. (2019). DeepLOB: Deep Convolutional Neural Networks for Limit Order Books. *IEEE Transactions on Signal Processing*, Vol. 67, No. 11, pp. 3001 – 3012.

Appendix A – Feature Definitions

All features are computed from YES-token quotes received at or before the decision time *t*. A quote state is valid when bid < ask, both lie strictly inside (0, 1), and the spread is at most 0.05. Missing lag values are encoded as zero together with a missing flag.

Table 19. The 18 features of row_hgb7. Features marked * are also used by the 14-feature V03 live model.

Feature	Definition
midpoint*	(best bid + best ask) / 2 of the current valid quote
logit_midpoint*	ln(m / (1 − m)) of the midpoint
spread*	best ask − best bid
source_age	seconds since the venue timestamp of the current quote
receive_age	seconds since the receive timestamp of the current quote
hour_sin, *hour_cos*	UTC hour of day on the unit circle
log_valid_updates30	log(1 + number of valid quote states in the last 30 min)
valid_state_fraction30	number of valid quote states divided by the number of recorded quote states in the last 30 min
delta5, *abs_delta5*, missing5*	signed and absolute midpoint change since t − 5 min; flag if unavailable
delta15, *abs_delta15*, missing15*	same over 15 minutes
delta30, *abs_delta30*, missing30*	same over 30 minutes

The activity logistic baseline uses midpoint, spread, log_valid_updates30, abs_delta5, missing5, abs_delta15, missing15, abs_delta30 and missing30. All logistic models standardise inputs and use L2 regularisation with C = 0.1.

Appendix B – Protocol Deviations and Failed Attempts

Every item below is preserved in the repository with its original outputs. None was rescued by changing a threshold after outcomes were seen.

Table 20. Deviations, failures and superseded versions, in chronological order.

Item	What happened	Handling
V3 activity gate	204,069 zero-count Kalshi rows; 19 markets missed activity minimum	V3.1 removed exactly those 19, label-blind; V3 failure kept
V3.1 holdout	Builder printed positive count (26,381) before consumption receipt	Disclosed; weights and formulas already frozen
Exact-price forecasting	Ridge 30.9% worse than persistence on MAE in a live audit	Reported as negative; target changed
V01 live runtime	Readiness 0/16: host clock faults	Frozen as failed; V02 built
June 8–14 capture	DNS outage failed 74 hours on both attempts	Outcome-blind refetch to separate directory; disclosed
V02 rehearsal 1, 2	Startup crash; WebSocket logged before clock was valid	Fixed with regression tests; archived
V02 activation 1	Memory about 317 MB/h and full rescans; 0 forecasts	Superseded by bounded inputs (510/510 equivalent)
V02 activation 2	Thread race forked hash chain; 0 forecasts	Single-writer journal; readiness re-run and passed
V02 live, first grid	Pre-loaded inputs not refreshed; 30 abstentions	Fail-closed; not patched during the test
V02 live, host stall	Concurrent job caused swapping; 4 grids missed	Heavy jobs deferred until close
V02 live, sleep/reboot	Lid-closed battery sleep 04:42 UTC, reboot 05:16 UTC, 27 Oct; source and emitter lost	Not restarted into the same run; not evaluable; continued as V02b (same frozen files, new run directory)
V02b live, sleep gap	Lid closed on AC 07:40–08:26 UTC, 27 Oct, before system-wide sleep disable	9 grids (270 decisions) scored as missed; processes survived; no restart, no backfill
V4 collector, same outage	Collector stopped by the reboot; checkpoint clean	Restarted from checkpoint with restart receipt; label-blind audit passed
Report rounding	June 1–7 interval printed as [0.039, 0.081]; artifact value 0.0805 rounds to 0.080	Caught by the reproducibility check; corrected
June 15–28 cohort v1	Only 15/60 contracts passed the end-date rule	Superseded before any price read by v2 (21 contracts)
V4 close-out audit	Frozen audit required run status ok; only 40% of 186,753 news rows had it, unlike the training data	Chain stopped. Disclosed v2 audit (any recorded status) hash-locked before consumption; chain then run once on 31 Oct
History-stack cohort	Frozen eligibility rules admitted 53 of a 150-contract target	Not loosened; smaller cohort reported

Item	What happened	Handling
History-stack capture	Frozen 32 MiB per-hour range limit exceeded in 134 of 336 hours (one timeout was retried and succeeded)	Not retried or recaptured; no forecasts issued in those hours; disclosed
History-stack evaluation	Panel built in chunks of 15 contracts to bound memory	Fixed in the frozen runner before capture; contracts are independent tapes
Segment confirmation selection	First selection attempt stopped when identifier and clock reads exceeded their frozen 64 MiB limit, before any economic column was read or any contract chosen	Attempt archived; a disclosed v2 runner with a 256 MiB limit selected the cohort
Segment confirmation capture	Versioned fetcher with a 192 MiB per-hour range limit (32 MiB in the history-stack test), frozen before selection	Disclosed; all 216 hours captured
Segment-aware stack window	Test period 22 July–8 August is the calendar period of the consumed V3.1 holdout; the archive ends on 10 August	User-authorized deviation, frozen before selection; V3.1, v1 and V4 markets excluded; no V3.1 data read
Segment-aware stack capture	One hour (31 July 20:00) timed out on the first attempt	Retried once under the frozen rule; all 432 hours captured
Baseline benchmark, TabICL	Its key–value cache ran at about 66 ms per row against 17 ms without it (timed on training rows)	Compute-only v2 amendment before any score: cache off; model, checkpoint and seed unchanged
Baseline benchmark, Sundial and Time-MoE	Added at the user's request after the protocol was frozen; model-hub code had first been refused by the sandbox	v3 amendment before any score: code read and hashed; scored on the nonoverlapping half-hour grid for inference cost; missing context values imputed
Baseline benchmark, end-to-end tier	TabM needed about 800 s for 9,212 rows on this host's CPU	v3 amendment: trained on a seeded 100,000-row subsample of the six blocks (seg_hgb used all rows; favours seg_hgb)
Baseline benchmark, TabICL prediction	Predicting 20,000 rows at once failed on the GPU (an 8.02 GiB buffer error); no output was written	Compute-only v4 amendment before any score: prediction in 1,000-row chunks; failed log kept
Development study v8	A further candidate (the online contract offset on seg_ref) was added at the user's request after the first run had started	The run was stopped before any fold finished, its log kept, and the candidate committed before the rerun

Appendix C – Reproducibility

Consistency check. verify_report_reproducibility_20261027 re-hashes the files locked by the cited release and data manifests (1,293 files in 94 manifests; two earlier locks pass only through their documented successor manifests) and fails if any is missing or changed. It checks the 105 headline claims listed in report/claims.json (each with its artifact, cohort, model and comparator) against their artifact values, and confirms that each claim's printed value still appears in the report source. It re-renders the figures from frozen outputs and checks that the PNGs are byte-identical, and with --recompute reruns the post hoc analyses. Numbers not listed in the claims file are not checked. It reads no holdout and starts no evaluation. Its first run, on 27 October, caught one rounding error in this report, now corrected. It was run again after close-out on 30 October and after the V4 and history-stack evaluations on 31 October and the segment and segment-aware stack evaluations on 1 November, and after the baseline benchmark and development study on 2 November, with and without --recompute, and passed each time.

All code is in the EventX repository. Frozen releases live under eventx/release/, and results with their hash manifests under data/. The principal entry points are listed below (run from the repository root with the project virtual environment). Commands marked once-only consume their block and refuse to run again; they are listed for audit, not for re-running.

Table 21. Principal scripts and outputs.

Purpose	Command or path
June 1–7 test (consumed, once-only)	python -m eventx.tasks.run_v7_june_replication_20261026
June 8–14 test (consumed, once-only)	python -m eventx.tasks.run_v7_new_identity_test_20261026 {select,capture,evaluate}
V03 secondary (consumed, once-only)	python -m eventx.tasks.evaluate_v7_new_identity_v03_20261026
June 15–28 test (once-only)	python -m eventx.tasks.run_v7_late_june_test_20261027_v2 {capture,evaluate}
Category breakdown	python -m eventx.tasks.analyze_v7_category_breakdown_20261027
Figures	python -m eventx.tasks.plot_v7_report_figures_20261027
Live runtime health (V02b)	python -m eventx.tasks.status_v7_forecast_runtime_20261026_v02b
Live evaluation (V02b, once-only)	python -m eventx.tasks.evaluate_v7_forecast_runtime_20261026_v02b
Close-out sequence (once-only)	zsh scripts/close_out_20261030.sh
V4 close-out (once-only, in order)	audit_v4_confirmatory_collection (stopped), then zsh scripts/run_v4_chain_v2_20261031.sh: audit v2, consume, prepare, build, evaluate
History stack: development	python -m eventx.tasks.develop_v7_history_stack_20261031
History stack: test (once-only)	python -m eventx.tasks.run_v7_history_stack_test_20261031 {select,capture,evaluate}
Segment discovery and replication (post hoc)	python -m eventx.tasks.analyze_v7_segment_replication_20261101 [features]
Segment confirmation test (once-only)	python -m eventx.tasks.run_v7_segment_confirm_test_20261101_v2 {select,capture,evaluate}

Purpose	Command or path
Market-state breakdown (descriptive)	python -m eventx.tasks.analyze_v7_segment_state_table_20261101
Segment-aware stack: development	python -m eventx.tasks.develop_v7_segment_stack_20261101
Segment-aware stack: test (once-only)	python -m eventx.tasks.run_v7_segment_stack_test_20261101 {select,capture,evaluate}
State-of-the-art baselines: feature rebuild (read-only)	python -m eventx.tasks.rebuild_v7_sota_features_20261101 {jul_13_21,jul22_aug8}
State-of-the-art baselines (post hoc, scored once)	python -m eventx.tasks.benchmark_v7_sota_baselines_20261101_v2 {tier_a,tier_b_inputs}; .venv-ts: benchmark_v7_sota_baselines_20261101 tier_b_forecast and _v3 tier_b2; _v3 {tier_c,evaluate}
State-of-the-art baselines: label-free preflight	python -m eventx.tasks.preflight_v7_sota_baselines_20261102 (must pass before the single evaluate)
Development study v8 (Section 6.11)	python -m eventx.tasks.develop_v8_hawkes_posterior_20261102
V4 audit revision (disclosed)	eventx/release/v4/AUDIT_REVISION_V2_20261031.md
Runtime tests	python -m eventx.tasks.check_v7_forecast_runtime_20261026_v02 (32 tests; V02 code, reused unchanged by V02b)
News bound (Section 4.3)	python -m eventx.tasks.analyze_v3_1_news_mde_20261027
Booster drivers (Section 6.4)	python -m eventx.tasks.analyze_v7_booster_drivers_20261027
One-command check	python -m eventx.tasks.verify_report_reproducibility_20261027 [--recompute]
Claim-to-evidence map	report/claims.json
Result formatter (read-only)	python report/fill_pending_20261030.py
Post close-out checks and hash manifest	zsh scripts/post_close_out_20261030.sh